High-resolution, early Paleozoic (Ordovician-Silurian) time scales

Peter M. Sadler†
Department of Earth Sciences, University of California, Riverside, California 92521, USA

Roger A. Cooper†
GNS Science, P.O. Box 30368, Lower Hutt, New Zealand

Michael Melchin†
Department of Earth Sciences, St. Francis Xavier University, Antigonish, Nova Scotia B2G 2VG, Canada

ABSTRACT

For much of the geologic time scale, resolving power appears to be limited by the duration of biostratigraphic zones and subzones. Yet these zones exploit the appearance and disappearance events of only a small fraction of the species they span. Computer algorithms can sequence an order of magnitude more events and make explicit the uncertainties that arise when calibrating the resulting time line. An illustrative case history builds an Ordovician and Silurian time scale from the geologic record of the entire graptolite clade—over 1900 species from more than 400 localities and sections worldwide. The same approach can be applied to any stratigraphic interval with detailed biostratigraphic observations. It provides the foundation for time scales and paleobiologic time lines.

Using logic similar to graphic correlation, optimizing algorithms search for a composite sequence of species first- and last-appearance events that minimizes the implied shortcomings of all the field observations. The algorithms minimize the number of species coexistences that are implied, but never observed, and the net adjustments needed to bring local range charts into agreement with a single composite sequence of events. After total section thicknesses have been rescaled to help normalize for variation in depositional rate, mean stratigraphic thicknesses are used to scale the intervals between adjacent events in the composite sequence. The resulting scaled composite sequence is converted to a relative geologic time scale by identifying stage and zone boundaries within the sequence of graptolite events. This relative scale is, in turn, calibrated by dated volcanic ash beds that were incorporated in the search for the optimal sequence. These dated events are also used to test for linearity of the scaled composite.

The final time scale has a potential resolving power of 0.02–0.1 m.y., more than ten times better than can be achieved by traditional zones. Graptolite zones vary widely in duration from as short as 0.1 m.y. to nearly 5.0 m.y . The mean duration of zones or zonal groupings calibrated here is 1.44 m.y. in the Ordovician and 0.91 m.y. in the Silurian. The average uncertainty in locating zone boundaries in a single composite sequence is about one-fourth of the mean zone duration. The variance resulting from differences in time scales developed from differing numbers of field observations and in response to changes in the optimization criteria gives a very conservative measure of the overall robustness of the method. These differences indicate an average uncertainty in the age of graptolite zone boundaries that is more nearly equal to the mean zone duration. For zone duration, the mean uncertainty amounts to about one-half of the length of an average zone.

Keywords: time scale, correlation, graptolite, Ordovician, Silurian, biostratigraphy, chronostratigraphy, resolution, calibration, uncertainty.

INTRODUCTION

Although ongoing changes in Earth systems may best be understood against the full scope of geologic time, the traditional subdivisions of the geologic time scale offer much less resolving power for deep time than the recent past. Rates of rapid global change are difficult to quantify for the early Paleozoic, for example, and prospects seem marginal for precise comparison of Pleistocene and late Ordovician glacial dynamics. The Ordovician time scale uses graptolite zones whose average durations must exceed 1 million years. Behind the set of named Ordovician zones, however, lies much more information about the appearance and disappearance of hundreds of graptolite species. Rather than working with the small number of derived biostratigraphic zones and stages currently used for interregional correlation, modern optimization algorithms make it possible to use the many hundreds of first and last appearances of fossil taxa in all available stratigraphic sections. This approach can increase resolving power of deep time scales by an order of magnitude or more and provides detailed time lines for paleobiologic studies.

A time scale that calibrates Ordovician and Silurian stage boundaries, using a time line of 1170 graptolite taxa based on 198 stratigraphic sections, was adopted by the International Geologic Correlation Project 410 (Sadler and Cooper, 2004) and the International Commission on Stratigraphy (Cooper and Sadler, 2004; Melchin et al., 2004). In neither of these publications was it fully explained how the calibration was implemented. This paper provides that explanation, shows the results of increasing the database to more than 400 sections and over 1900 taxa, and extends the interpolation of ages down to zone boundaries in order to explore the potential and practical limits to resolving power.

First, we define some terms and formalize the time-scale problem. Then, turning to the construction of composite time lines, we draw rules for sequencing bioevents from previous methods that either amalgamated many taxon range charts by graphic correlation (e.g., Shaw, 1964; Murphy and Edwards, 1977; Kleffner, 1989, 1995; Cooper 1992, 1999; Armstrong, 1995; Sweet, 1995; Sweet and Tolbert, 1997; Sweet et al., 2005) or sequenced isolated faunas (e.g., Guex, 1991; Alroy, 1992, 1994). The rules are applied to the large Ordovician to early Devonian graptolite data set using optimization techniques developed to solve famously hard

†E-mail: peter.sadler@ucr.edu; r.cooper@gns.cri.nz; mmelchin@stfx.ca

GSA Bulletin; May/June 2009; v. 121; no. 5/6; p. 887–906; doi: 10.1130/B26357.1; 12 figures; Data Repository item 2008252.

sequencing tasks, such as the traveling salesman problem (Bellmore and Nemhauser, 1968). The salesman's problem is to visit all of a large number of cities once, and only once, in the sequence that yields the shortest total travel distance. For the time-scale problem, the cities become species appearance and disappearance events; the travel distance becomes the misfit between the positions of events in a chosen sequence of events and in the individual stratigraphic sections. For both problems, the number of possible sequences is enormous. The available information for our Ordovician-Silurian time-scale problem, though voluminous, is still insufficient to uniquely sequence so many graptolite appearance and disappearance events. We use this fact to develop estimates of three conceptually different kinds of uncertainty of stratigraphic position in a time line.

Definitions and Terms

An *event* is any geologic event whose stratigraphic record can be used in time correlation. An *event level* in a local stratigraphic section of sedimentary strata (simply termed a *section,* hereafter) is the horizon of the same age as the event; it may fall within a depositional hiatus. *Dated events* are those whose ages have been reliably determined by radioisotopic methods. *Biostratigraphic events* are the first and last appearances of a fossil taxon in a local stratigraphic section; these are range-end events. *Range* refers to the stratigraphic range of a species. The *local range* is the span of time or rock thickness in which a fossil species is found in one section. The combined record of multiple locations leads to a *composite range*, the ends of which are estimates of two global events—the first appearance datum (*FAD*) and last appearance datum (*LAD*). *Range extension* refers to an increase in the length of an observed local range necessary to match the composite range. The ordering of all biostratigraphic events into a single ordered time line that best fits the observed order in the local sections results in an *ordinal composite sequence* of events (or *best-fit composite*). Formulae for computing the misfit between a composite sequence and the field observations are *objective functions* (also known as cost or penalty functions).

When all pairs of events in the ordinal composite have been spaced proportionally to the inferred (or calculated) time intervals that separate them, the result is a *scaled composite*. A *relative geologic time scale* (or relative scale) is a scaled composite in which the positions of conventional stage and/or zone boundaries have been located. When the ages in millions of years have been determined for stage or zone boundaries, the result is a *calibrated geologic time scale*, or geochronologic scale. Other terms are italicized where they are defined in the text.

The Nature of the Time-Scale Problem

Every fossil taxon generates at least two unique biostratigraphic events—its global first and last appearances in the fossil record (FAD and LAD). An ordered succession of these events provides an ordinal scale for geologic time but does not reveal the duration of the time intervals between events. Age control for a true interval time scale derives from dated events. Single sections of sedimentary strata preserve only a fraction of all the biostratigraphic events and dated events for a given time interval. Furthermore, in each section the observed taxon ranges are almost inevitably shorter than the global range. The highest finds of a fossil taxon in a local section most likely fall below the horizon that corresponds in age with its LAD, and the local lowest finds fall above the FAD. These are straightforward consequences of ecology, biogeography, and fossil preservation: species appear at unique times and places; their subsequent geographic range varies through time; the living organisms are patchily distributed within that range; sediment accumulates discontinuously; few individual organisms are buried and fossilized at any one site; and many fossils are recovered in unidentifiable condition. Thus, real differences in the local timing of appearance and disappearance events are confounded with artifacts of preservation and collection.

The differences between observed local range ends and the corresponding FADs and LADs vary from place to place. Thus, range charts from individual sections can contradict one another in detail concerning the apparent sequence of species appearance and disappearance events. The composite record of many locations can compensate for local incompleteness, and the ends of composite ranges will better approximate the global FADs and LADs. Consequently, the following task captures the essence of the paleontologic time-scale problem.

Merge the dated events and contradictory successions of biostratigraphic events from many local thickness scales (the measured sections) onto a single time line that correctly orders the paleontologic events and scales all the time intervals between them.

It is useful to distinguish between the compositing and the interpolating parts of the time-scale problem. *Compositing* merges the contradictory, ordered subsets of events from different localities into one overall ordered sequence—the ordinal composite sequence. The computational challenge of compositing increases dramatically with the quest for increased resolving power. Higher resolution requires more closely spaced events, which, in turn, increase the likelihood of contradictions in observed order. In addition, the number of possible composite sequences increases factorially with the number of events included.

The *interpolation* task estimates ages for paleontologic events according to their position relative to dated events. Interpolation introduces simplifying assumptions to convert the ordinal composite to an interval scale, or *scaled composite*. For this reason interpolation is often best undertaken after the compositing process has identified the best-fit sequence of events without recourse to assumptions about the steadiness of rock accumulation or biologic change. Where there are sufficiently numerous dated events, however, interpolation can be attempted prior to compositing. In the late Cenozoic ocean cores, for example, the same fossiliferous strata that record the local taxon ranges may also preserve paleomagnetic and isotopic events with well-established ages. Each core might support local interpolation of age for the paleontologic events on the local thickness scales (Berggren et al., 1995). Although such interpolations assume steady accumulation and no hiatuses, the interpolation distances are short and the associated error can be reduced while compositing independent estimates from different sites. Cenozoic land mammal faunas present a less favorable case (Alroy, 1992, 1994). Long, fossiliferous, measured sections are few; most information derives from isolated faunas; and the composite sequence is generated primarily by evidence of coexistence. Dated events are sufficiently abundant, however, to allow a nonlinear interpolation on the composite sequence.

Note that the calibration of several late Cenozoic FADs and LADs is so advanced that it is tempting to treat them as dated events. The locally preserved highest and lowest finds typically need adjustment, however, up and down section, respectively, in order to reach horizons of the same age as the true LAD and FAD. The principle that FADs and LADs should be estimated by minimizing these adjustments is fundamental to graphic correlation ("economy of fit," Shaw, 1964) and permits optimal composite sequences to be found for Paleozoic paleontologic events.

Traditional Approaches to Paleozoic Time-Scale Problems

The potential resolving power of marine paleontologic history is probably not markedly worse in the Paleozoic than the Cenozoic. Rather, the subduction and accretion of Paleozoic ocean floors has left a more limited, discontinuous, and disturbed sedimentary record. Seafloor spreading rate cannot be used to calibrate Paleozoic reversals in the geomagnetic field. Paleozoic dated events are relatively rare and often established outside the fossiliferous sedimentary sections for which detailed taxon ranges have been determined. Consequently, the compositing task cannot be completed against a time scale. Instead, the paleontologic events must first be placed in order, then spaced on a relative time scale, and finally calibrated by interpolation between the dated events. This progression of steps completes first those tasks with the fewest inbuilt assumptions.

No composite sequence of events can agree with all the local section range charts; the compositing task simply minimizes the net misfit. Traditional biostratigraphic procedures systematically cull out the most conflicted events leaving a much smaller number of *index events* that are preserved in an internally consistent order in many different sedimentary sections ("homotaxy," Scott, 1985) and these events are used to define the boundaries of biozones. Once correlated from place to place, the index events complete the compositing task, but at a price paid in loss of resolution. The culling process must eventually lead to homotaxy: when so few taxa remain that their ranges do not overlap in time, then the local underrepresentation of range length can no longer cause index range-end events to be preserved in contradictory order.

Typically, ages are interpolated only for epoch or stage boundaries. If each of these units contains several dated events, then the boundary ages can be interpolated by error analysis on a true time scale (Harland et al., 1990, chapter 5). More often, dated events are insufficient and interpolation is performed by "graphic methods" (Odin, 1985) upon some relative time scale that estimates the relative durations of the units. Dated events are loosely located within biozones and the duration of zones must be adjusted by trial and error to achieve a linear regression of age on stratigraphic position (e.g., McKerrow et al., 1985). Different adjustments can lead to significantly different time scales with the same control points and equally good linear regressions (Gale, 1985).

All time scales derived by these relative methods require at least one of the following assumptions: (1) that sediment accumulated steadily between the dated event horizons with no hiatuses or condensed intervals (e.g., Churkin et al., 1977); (2) that accumulation was so regularly periodic that successive sedimentary cyclothems represent time intervals of equal duration (e.g., Kominz and Bond, 1990); or (3) that the evolutionary rate is sufficiently regular that the number of biostratigraphic events is a guide to elapsed time (e.g., Van Hinte, 1976; Westermann, 1984; "biostratigraphic discrimination" of Harland et al., 1990). Although none of these assumptions is likely to be true, they cannot be avoided altogether if a relative scale is needed for interpolation. Unless applied judiciously and explicitly, they surely compromise the interpolation task (Smith, 1993).

The relative duration of Paleozoic stages has been estimated, for example, by comparing the number of zones (i.e., index events) they contain (Cocks, 1971; Boucot, 1975; Ziegler, 1978; McKerrow et al., 1980). The implicit assumption that zones do not differ much in duration is at odds with estimates of the relative duration of zones from their thicknesses in a section of uniform black shales (Churkin et al., 1977; Carter et al., 1980). Cooper (1992, 1999) estimated the relative duration of graptolite zones from shale thicknesses in composite sections built by graphic correlation. His composite sections, constructed independently from black shale in Scandinavia and Newfoundland, resemble one another to a striking degree and imply that early Ordovician graptolite zones vary considerably in duration. We explain below how to: (1) extend Cooper's approach to a much larger graptolite database; (2) migrate the concept of "economy of fit" from manual graphic correlation to computer algorithms that can manage large data volumes (Kemple et al., 1989, 1995); and (3) add some advantages from consideration of taxon coexistences (Guex, 1991; Alroy, 1994; Sadler, 2004).

GLOBAL GRAPTOLITE DATA BASE

Graptolites are ideal fossils for long-range correlation. They formed a major component of the early Paleozoic plankton and are found in a wide range of sedimentary facies. They are most abundant and diverse in sediments of the outer continental shelf, slope, and base-of-slope (Finney and Berry, 1997; Cooper, 1998; Chen et al., 2001). Many graptolite species are widespread globally, and many have relatively short time ranges. Together with conodonts, they are the primary fossil group for global correlation of Ordovician and Silurian stratigraphic sequences.

Currently our graptolite database comprises 17,861 locally observed range-end events for 1983 taxa, and 131 local observations of 57 other events (23 dated events and 34 marker beds) in 446 sections worldwide (Fig. 1). They cover the time interval from earliest Ordovician to early Devonian. The information for the sequencing task amounts to 15,126 range ends that do not coincide with the ends of a measured section,

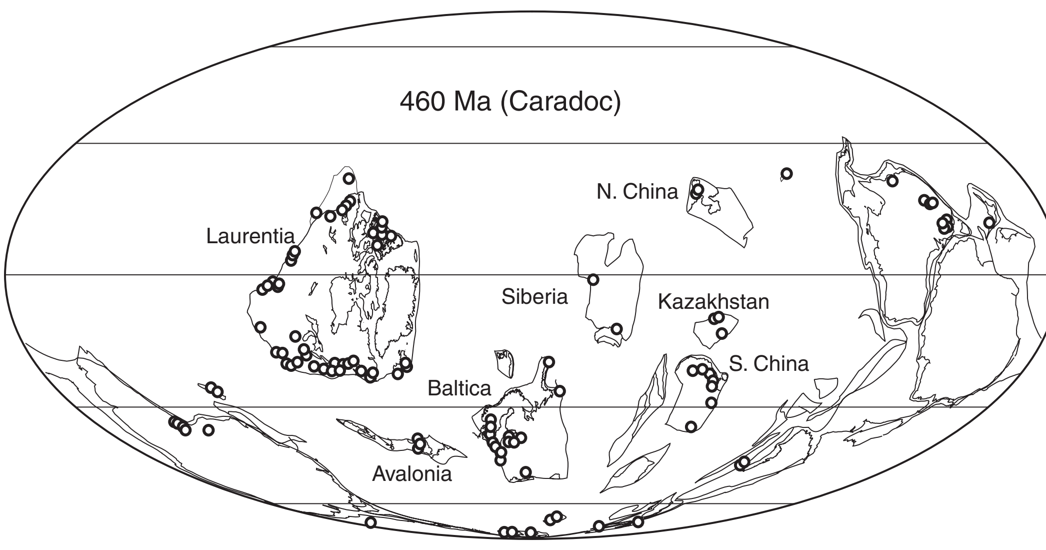


Figure 1. Global distribution of sites that contain one or more graptolite-bearing sections used in this study, plotted in a best approximation of their paleogeographic position on a 460 Ma terrane map (after Cocks and Torsvik, 2002).

27,464 different pairs of taxa that are observed to coexist (range overlap), and 242,464 different pairs of first- and last-occurrence events, for which the first-occurrence was observed before the last-occurrence in the same section.

The data sources, database software, and sequencing program are archived as supplementary data in the GSA Data Repository[1] with discussion of the selection criteria and reference manuals for the sequencing software. We attempted a total-data approach using all published range charts that meet modern standards of graptolite taxonomy. Twelve regional zonal range charts (zonal composites) are included that use a scale of zones to summarize taxon ranges—Australasia, the Canadian Arctic, Wales (2), Scotland (2), the Upper Yangtze and Jiangnan regions of China, East Siberia, Kazakhstan, Barrandia, and Poland. These are expert compilations that summarize the faunas from numerous localities not available to us as individually published range charts and faunal lists. The Barrandian chart, for example, integrates more than 50 localities. The number of localities summarized in the Australasian and British charts is estimated to be in the thousands.

For dated events we have chosen mostly U-Pb dates on zircon crystals in volcanogenic beds that are well constrained biostratigraphically, and analyzed by the thermal ionization mass spectrometric isotope dilution (TIMSID) method. This conservative selection follows the conventions adopted in recent papers using isotopic dates to refine and calibrate the Paleozoic time scale (e.g., Bowring et al., 1993; Tucker and McKerrow, 1995; Davidek et al., 1998; Tucker et al., 1990, 1998; Gradstein et al., 2004). For the reasons discussed by Cooper and Sadler (2004), ages determined by the sensitive high-resolution ion microprobe (SHRIMP) method are not used here.

Only three of the dated beds lie within sections for which detailed graptolite range charts have been established on the basis of numerous fossiliferous horizons. For the others, the position of the dated bed is constrained by its position relative to one or two graptolite or conodont faunas. For dates associated with conodonts, a graptolite with a comparable range was selected, guided by correlations between the graptolite and conodont zonations (Bergström, 1986; Bergström and Mitchell, 1992; Bergström and Wang, 1995; Löfgren, 1996; Cooper, 1999). Dated beds that cannot reliably be linked to a graptolite were not used. Thus, our calibration lacks some dates used in other time scales.

COMPOSITING, SCALING, AND INTERPOLATION

The calibrated time scale is developed in four steps (Fig. 2). Step 1 builds a purely ordinal composite sequence of paleontologic and dated events from the raw field data. Step 2 scales the intervals between events using assumptions about rates of rock accumulation and biologic turnover. Step 3 locates standard zone boundaries in the composite sequence using conventional index events. Step 4 interpolates the ages of zone boundaries from the dated events to calibrate a time scale. The four-step process is repeated after enlarging the database or changing the compositing rules. This leads to multiple solutions whose differences are a measure of the level of uncertainty that results from insufficient constraints.

Step 1: Building the Ordinal Composite Sequence

All stratigraphic events are arranged into an order that best fits the information from all local range charts and faunal lists. This enormous computational task is undertaken using an optimization algorithm (Kemple et al., 1995; Sadler and Kemple, 1995; Cooper et al., 2001) based on an elegant analogy with the growth of a perfect crystal by annealing and slow cooling (the "simulated annealing" heuristic of Kirkpatrick et al., 1983). The sheer amount and variety of stratigraphic information requires this kind of efficient search algorithm, which can proceed by trial and error and does not need to consider all imaginable sequences of events (Sadler and Cooper, 2003). The algorithms implement familiar, straightforward biostratigraphic best practices, and the input data are actually very simple. Each locally observed stratigraphic event generates one line of input data (Fig. 3) that describes the event type and its observed position in the section.

Two kinds of relationship between events are extracted from the input data: *pairwise ordering of events* (the stratigraphic order of all pairs of events that can be observed in the same local section) and *pairwise separation of events* (the stratigraphic distances separating the members of each pair). These relationships may differ from section to section. Nonetheless, they serve to eliminate nonfeasible composite sequences of events (i.e., they constrain the search to feasible sequences) and to guide the search for those sequences that best fit the field data (i.e., optimization). In this sense, the search for the solution to the time-scale question posed above proceeds as a *constrained optimization*, using the program Conop9 (from CONstrained OPtimization for the Windows 9x-NT-XP operating system; Kemple et al., 1995; Sadler et al., 2003).

Observed pairwise orderings of events provide powerful *constraints* on the number of composite sequences that are feasible, because we assume that the observed local ranges cannot be longer than the true global ranges (i.e., the data are free of the results of reworking or borehole caving). Thus, if the first occurrence of one species is observed to lie below the last occurrence of another in any local section, range extensions cannot reverse this order. Any feasible composite sequence of FADs and LADs must include this pairwise order. The observation that two taxa coexist amounts to two of these pairwise orderings: both first-occurrences must precede both last-occurrences.

Pairwise ordering can also drive the optimization process. Composite sequences that include large numbers of events and combine data from many paleocontinents almost inevitably imply more taxon coexistences and FAD-before-LAD pairs than have been confirmed by field observations. Several correlation programs ("Biograph"—Guex, 1991; Savary and Guex, 1999; "Conjunct"—Alroy, 1992; "Appearance Event Ordination"—Alroy, 1994) build composite sequences of taxon appearances and disappearances simply by minimizing the number of these unobserved pairwise orderings.

Algorithms that use pairwise separation consider the stratigraphic distance between observed events. They minimize the sum of all adjustments needed to bring every local sequence of events into agreement with the composite (Kemple et al., 1989, 1995). The permitted adjustments are extensions to the observed stratigraphic ranges of taxa (*range extensions*), which may be measured in several ways (*misfit measures*). Graphic correlation (Shaw, 1964; Hood, 1986) uses rock thickness to measure the size of range extensions. By seeking solutions that minimize these extensions, graphic correlation tends to favor sequences of events that match those observed in thick sections where event levels are widely separated. Our global database contains sections with substantial differences in sediment accumulation rate, and the thicker sections are not necessarily those with the most complete records of graptolite sampling or highest species diversity. We prefer to favor sequences observed in the most taxon-rich and intensely sampled sections. These sections have the most event levels, for a given time span. To favor them, we measure the size of a range extension by counting the number of event levels leapfrogged by the exten-

[1]GSA Data Repository item 2008252, Additional explanation of procedural details; tables of radioisotopic dates and zone calibrations; list of sections and sources; manuals and Windows executable files for sequencing software and data management (conop9.exe and conman9.exe), is available at http://www.geosociety.org/pubs/ft2008htm or by request to editing@geosociety.org.

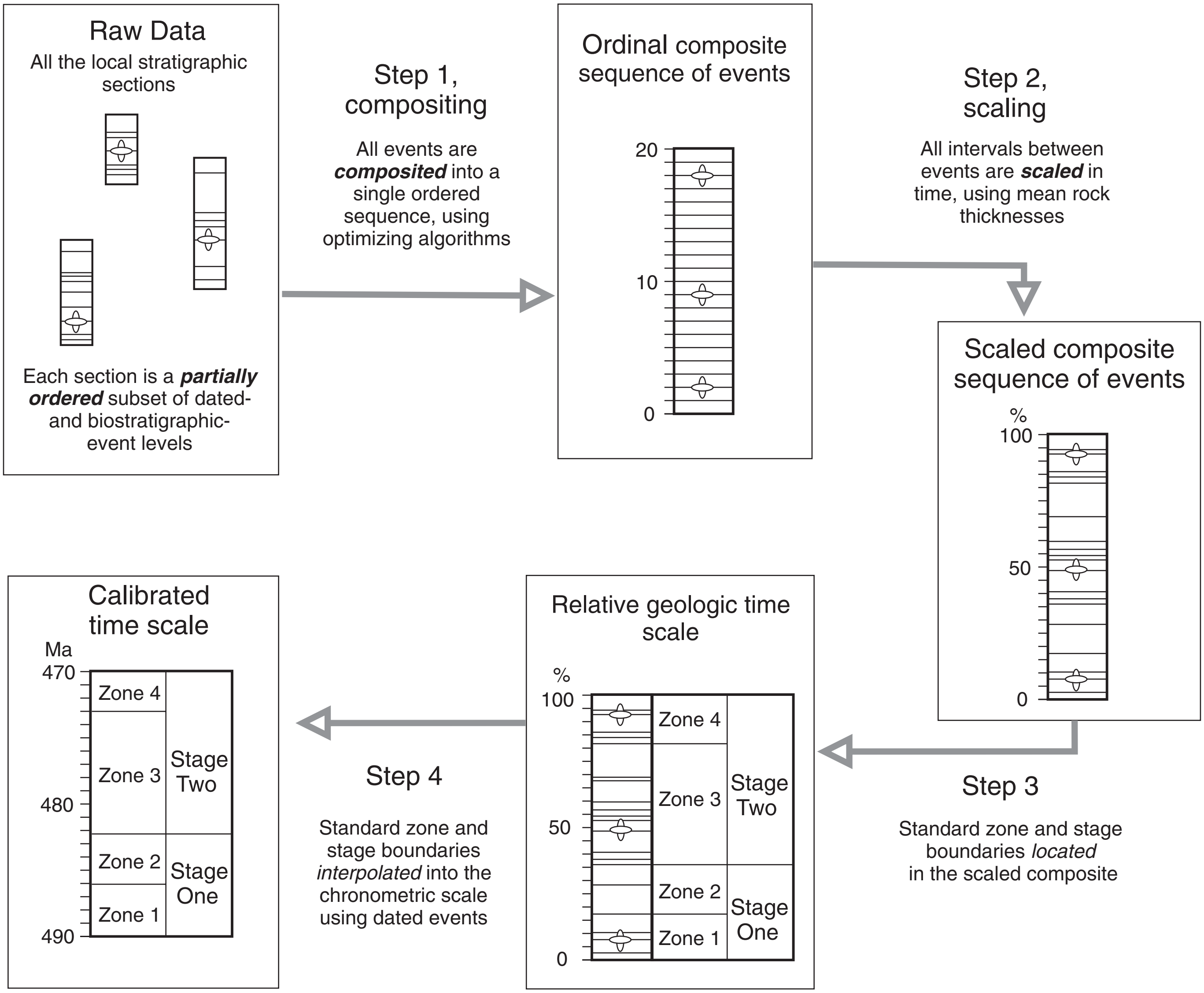


Figure 2. Flow diagram to show the steps and terms used here for procedures in building the time scale. The schematic illustrations show the positions of paleontologic events (fine horizontal lines) and dated events (orbital icons) as arrayed in raw field observations, an ordinal composite sequence, and a scaled composite sequence, in which traditional zone boundaries may be identified and numerical ages interpolated.

.	.	.	.	. . .	2	1.00	1.00
33	2	11	22.1	8	2	1.00	1.00
33	2	17	340.0	11	2	1.00	1.00
34	4	11	22.1	8	0	1.00	1.00
35	5	17	443.3	14	0	442.3	444.3
36	1	21	18.5	10	1	1.00	1.00
36	1	22	66.0	32	1	1.00	1.00
36	1	2	-9.57	5	1	1.00	1.00
36	2	21	24.1	14	2	. . .	. .

Figure 3. An excerpt from a CONOP input file. Each locally observed event is one 8-column line that records event number, event type, section number, height, level, movement code, and two weights. The line in the dashed box reads as follows: "taxon number 36 appears (type 1 event) in section 21 at a height of 18.5 meters, which is the 10th event horizon from the base of the section; this appearance event may only be adjusted downsection (code 1) and with a relative weight of 1.0; were upsection adjustment not prohibited, it too would be weighted 1.0." Last occurrence events (shown by type 2 in column 2) may be adjusted downsection only (code 2 in column 6). Marker beds (events of type 4, column 2) and dated beds (type 5, column 2) may not be adjusted up or down (code 0, column 6). For dated events, the weights are replaced with the minimum and maximum expected age values (e.g., mean ±2σ).

sion. This strategy is not new; it is adopted from Edwards' (1978) no-space graph method.

The Conop9 program was configured to implement two optimization strategies. Both are constrained by having to honor all observed first-appearance–before-last–appearance-orderings and minimize the additional number of such pairs in the composite sequence. One strategy (the "SEQUEL" objective function in Conop9) uses only this information. The other minimizes net taxon range extensions as measured in event levels (the "LEVEL" objective function in Conop9). The latter is slow but preferred because it uses more information. Both strategies adopt the Conop9 "sensu-stricto" requirement for evidence of pairwise ordering; this requires that the two ranges have been observed to overlap and not merely coexist in one collection or level. This is necessary because any single graptolite collection is potentially time-averaged, and co-occurrence of species in a sample does not ensure that their ranges actually overlapped in time. It also allows use of regional composites in which taxa are grouped by biozone: each zone can be treated as a time-averaged collection without implication that all taxa in a zone have overlapping ranges.

The operational details of the Conop9 search process, fully explained in the references given above and in the supplementary materials (see footnote 1), are briefly summarized here. The search starts from a random sequence of events that must honor all first-before-last appearance observations. This is achieved by placing all FADs in random order at the beginning of the starting sequence and all LADs in random order at the end. Thus, initially, all taxon pairs coexist. Other events (marker beds and isotopically dated beds) are placed after the FADs, before the LADs, and in an order that respects any superpositional relationships and numerical ages. Pairs of dated events that overlap at the 2σ level may appear in either order.

The (very large) misfit between the raw stratigraphic sections and the starting sequence is calculated according to the objective function chosen for the search. The determination of misfit based on range extensions requires finding the smallest set of adjustments that will bring the order of events in each section into agreement with the composite. These adjustments may lengthen observed taxon ranges, but the local position of marker beds and dated beds may not be altered. Thus, the inclusion marker beds and dated beds adds considerable constraint and efficiency. Note that the correct order of two dated beds is determined on a scale of numerical age, whereas the correct order of dated beds and a FAD or LAD is determined by superposition. A first appearance cannot be adjusted upward, for example, and the corresponding FAD must be older than any dated bed observed higher in the same section. The two measurement scales (time and thickness) do not present a problem because the relationships are constraints that immediately identify impossible sequences (infinite misfit); they are not measures of fitness used to locate the best of the possible solutions that remain.

The initial sequence is then subject to a very long series of mutations, each of which chooses one event at random and moves it to a new random position in sequence. If the new sequence fails to meet all constraints, the mutation is abandoned, and another is tried. Otherwise the misfit is recalculated for the mutated sequence. If the fit has improved, the new sequence is adopted, and the next mutation is applied to it. If the fit has been made worse, the mutation is adopted or rejected according to a probabilistic rule (Kirkpatrick et al., 1983) that is more likely to accept deleterious mutations early in the search and if the increase in misfit is small. By this relatively simple process, the search progressively finds better and better solutions until, eventually, it can make no further improvement.

The tolerance for accepting deleterious mutations prevents the search from becoming "trapped" at a suboptimal sequence. The search is said to cool progressively as the formula for this tolerance (based on the Boltzmann equation of physical chemistry) reduces the frequency of accepting deleterious mutations and the process becomes progressively more discriminating. A suitable cooling rate must be determined by experience. The whole process can be repeated with different cooling strategies until it is clear that no further improvement can be reached. We use a conservative cooling regime (the "autocooling" option in Conop9) that proceeds to a lower temperature only after reaching diminished returns (i.e., no further improvement in tens of thousands of mutations) at the previous level. For our largest data sets, 10 million or more mutations are required to converge on an optimal solution. These long runs take 2–15 days on desktop computers, depending upon the objective function and the processor speed. Nevertheless, there may still be fragments of the best-fit sequence that biostratigraphers recognize as nonoptimal. The number of events is too large and the amount of information insufficient to eliminate all anomalies.

It is a common misconception that sequencing programs, such as Conop9, treat all species as if they were equally useful for correlation. In fact, the optimal sequence is automatically most heavily influenced by those taxa that are seen frequently and in a consistent order. Expert biostratigraphers apply the same criteria manually. The quality of results obtained when including all taxa with equal weight proves that the biostratigraphic method is less subjective than some might imagine (Kemple et al., 1995).

Traditional graphic correlation was designed to combine the record of stratigraphic sections that overlap substantially in age—it minimized only the range extensions. In our data set, the individual sections span much less time than the composite sequence, and the time spans of many pairs of sections do not overlap at all. In this regard, our instance of the time-scale problem resembles those that have only isolated faunal samples. Such instances can be solved by consideration of excess coexistences alone. Similarly, the inclusion of excess FAD-before-LAD pairs in our objective functions improves the efficiency with which the algorithms stack up nonoverlapping sections—it allows faunal dissimilarity to become an indication that sections might be of entirely different age.

Taxon range-end events that occur at the upper or lower limits of a local stratigraphic section are most likely to be artificial range truncations and lack useful information. Because range extension carries no additional cost once a range end reaches the limits of a section, these events automatically contribute nothing to the misfit measures, and it is not necessary to delete them from the data set. Similarly untrustworthy clusters of range ends may occur directly below or above a significant hiatus. For this reason, we split some sections in two at known surfaces of hiatus. The known order of the separate parts is retained, however, by treating the lower and upper bounding surfaces of the hiatus as two marker beds of known superpositional order; the lower surface forms the top of the older section, and the upper bounding surface forms the base of the younger section. Although this ploy prevents the local hiatus from misinforming the composite sequence, events that lie within the interval of hiatus cannot be resolved unless other sections contain a fossiliferous rock record of this time. Although composite sequences offer the best opportunity to mitigate the impact of hiatuses, and our total-data approach has assembled the largest data set we could find, there are surely unfilled hiatuses that remain. Not all of these will be trivial gaps in the composite sequence.

Step 2: Scaling the Composite Sequence

The result of Step 1 is an ordinal composite with no implied sizes for intervals between each event. In Step 2, intervals between events in the composite sequence are scaled according to the relative stratal separation of events in all the measured sections, *after* the observed taxon ranges in the sections have been extended and missing taxa inserted, to match the optimal composite sequence (Fig. 4). Average rescaled intervals between events are calculated in two objective and reproducible steps that attempt to mitigate the impact of unsteady and nonuniform accumulation rates.

First, the total thickness of each section is rescaled to the number of events that the section spans, in the composite sequence. This utilizes long-term faunal turnover to provide a crude initial estimate of the relative duration of the time intervals represented by whole stratigraphic sections. Sections that span the same series of biologic events do span the same time interval and rescale to the same thickness by this process. The oversimplification arises from equating the number of events in the series with duration, regardless of age. Nevertheless, the simplification becomes more palatable for long intervals and seems better than assuming that all sections accumulated at the same long-term rate. The relative rock thicknesses between event levels retain their original proportions in each rescaled section.

The shorter time intervals between successive paleontologic events certainly are not equal. In the second step, the interval between adjacent events in the scaled composite is taken to be the mean of the rescaled thicknesses between the two events in all sections that span them both. The mean includes the zero values that arise when multiple events are placed at the same local horizon. Ties in the placement of events (irresolvable clusters) were not permitted in the unscaled ordinal composite, but they are recognized in the rescaled sections and the scaled composite sequence. The pace of biologic change is surely not steady. During the Hirnantian, for example, extinction rates rose above background levels; this results in numerous LAD events whose spacing is zero (irresolvable) in the local sections. Including these zero values in the averaging process is an attempt to mitigate the effects of intervals of unusually high turnover. Intervals of unusually low turnover may lead to underestimates of elapsed time. There is no way to avoid all questionable simplifying assumptions for the scaling and interpolation tasks; step 2 simply keeps them minimal, explicit, and reproducible.

Step 3: The Relative Geologic Time Scale

Because the scaled composite resembles an exceptionally complete stratigraphic section, it is a conceptually simple task to locate the levels of index events that define zone boundaries. In practice, this step is as troublesome as global correlation itself. Composite sequences built by correlation algorithms almost inevitably include some surprising placements of events. The surprises have four causes.

First, as a result of provincialism and migration, the chosen index events may differ from region to region or appear in a different order. Thus a composite sequence of true global ranges would appear different from any one traditional regional zonal composite. As with global correlation, a zone boundary may need to be identified by a cluster of appearances

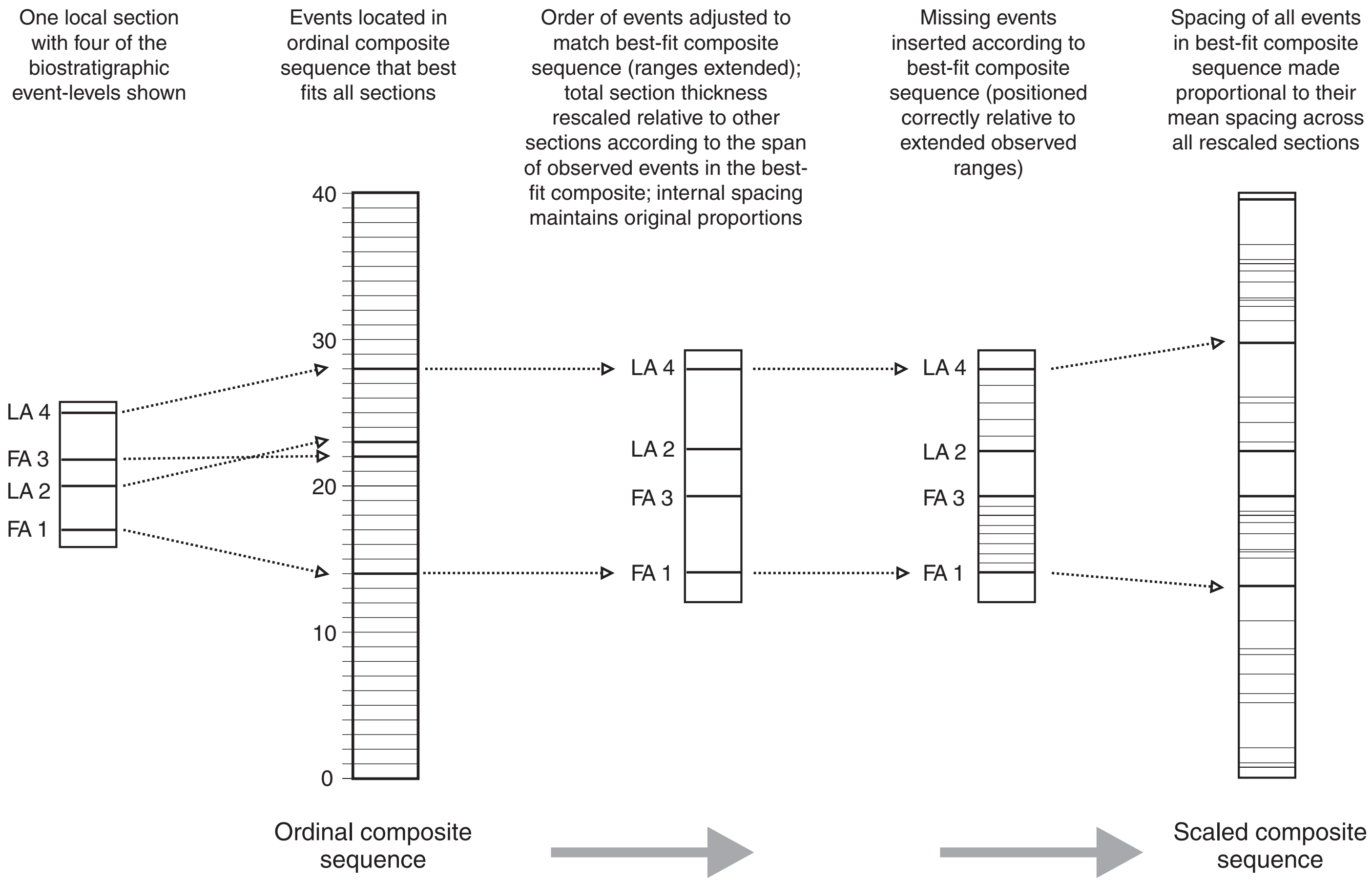


Figure 4. Schematic illustration of the construction of the scaled composite sequence from the ordinal composite and the spacing of events in the local sections. In the example, a section with four events is shown; FA—first appearance; LA—last appearance. See text for details of the procedure.

of taxa that characterize the zone in various regions. Some locally reliable index taxa might not fall within this cluster. One uncertainty in boundary placement, therefore, is the length of the interval over which the majority of locally characteristic taxa appear.

Second, taxonomic practice is not always uniformly precise and standardized. Some composite taxon ranges are too long because they incorporate a mixture of taxonomic practices. By default, the data will surely include some composite species concepts. Third, even with uniform taxonomy, taxon ranges may simply be too poorly constrained. Some taxa are too rarely found, and taxa known only from finds in the highest or lowest samples in a section are scarcely constrained at all. We include all taxa, however rare, in order to avoid arbitrary culling decisions and in the attempt to capture global rates of biotic change. Fourth, as mentioned above, the Conop9 trial-and-error search for the best ordinal composite sequence may be imperfect and almost inevitably includes some fragments of the sequence that are not optimal.

Most graptolite zones are based on first-appearance events of index taxa and correspond to interval zones (Salvador, 1994). In our scaled composite, some sets of zones could be located directly by finding their definitive index events—commonly the first appearance of the taxon for which the zone is named. This works best when graptolite species were relatively uniform around the world in their stratigraphic distribution, as in the late Silurian (Holland and Bassett, 1989). Biogeographic provincialism is stronger at other times, especially in the Ordovician, when many graptolite species have geographically restricted distributions (Bulman, 1971; Skevington, 1974; Berry, 1979). Furthermore, several species, such as *Extensograptus hirundo,* have stratigraphic ranges that appear to differ significantly and consistently from province to province. For these reasons it has not been possible to establish a suite of graptolite zones through the Ordovician with global application as has been done for the Silurian (Koren' et al., 1996). For example, one of the most reliable and widely applied Ordovician zonal suites, that of Australasia, which lay within the tropics in Ordovician time, cannot be applied in high paleolatitude regions such as Avalonia (Skevington, 1974). Nonetheless, we attempt to locate the Australasian zone boundaries in the scaled composite.

For the reasons just mentioned, it is necessary to rely upon more than the index taxon when locating zone boundaries. We place zone boundaries at the earliest appearance in the composite of a *cluster* of species (generally three or more) that are normally taken as characterizing the zone. Often this cluster includes the zonal taxon. Where the cluster does not include the zonal taxon, an alternative species is selected from the cluster as a zonal proxy (Table 1A in the supplementary materials; see footnote 1). The zones, as used here, are therefore closer to assemblage zones in nature (Salvador, 1994), but their boundaries are the best approximation to the boundaries as defined and used in Australasia.

Step 4: Calibration of the Time Scale

Dated events are placed in the composite sequence by the compositing process according to the graptolite or conodont species associated with them. They serve two purposes: to test whether the scaled composite is a plausible proxy for time (tests are discussed below) and to allow interpolation into the chronometric scale of numerical ages, leading to a calibrated time scale. In our data the dated events define a nearly linear regression when their positions in the scaled composite are plotted against their numerical ages (Fig. 5). This demonstrates the remarkable power of the optimizing and scaling procedure and that the scaled composite itself is a good proxy time scale. The relative *duration* of zones and stages can be determined directly from the relative time scale without further recourse to the dated events. The numerical ages of all events, including those that define zone and stage boundaries, as well as the durations of the zones in millions of years, can be derived from the scaled composite, via the regression function.

While compiling the data, over 10 yr, we generated ten scaled composite sections and fully calibrated four of them. Since publication of the time scales based on a composite that included 198 sections and 1207 taxa (Cooper and Sadler, 2004; Melchin et al., 2004), we have calibrated composites of 1446 taxa from 256 sections ("composite 256") and 1928 taxa from 430 sections ("composite 430"). Finally, from this largest data set we culled out half the sections leaving those that maximized geographic coverage, time span, and taxon richness, with a minimum number of taxonomist-authors ("composite 214").

The resulting calibrated geologic time scale is discussed below, but first we report the results of several tests of plausibility carried out during the process of building the scale and comment on the resolving power of the emerging time scale.

TESTS OF PLAUSIBILITY

The plausibility and robustness of our results were tested at six successive steps in the progression to a calibrated time scale. First, we examined the magnitude of adjustments to the observed ranges. Typical solutions fit all the local range charts to a single sequence with quite modest taxon range extensions of only 1–2 local event levels per observed event. Slightly more than half of all observed range ends require no adjustment at all.

Second, the ordinal and scaled composite sequences can be tested in terms of the correlations they predict for the stratigraphic sections (Fig. 6). The relative span of sections through the Ordovician and Silurian accords well with our expectation, based on their published durations, expressed in terms of stages and zones.

Third, the unscaled composite sequences can be compared with expert compilations of ranges by biozone. Figure 7A shows a strong correspondence of our ordinal sequence of events with VandenBerg and Cooper's (1992) compilation of the known ranges of 333 species in 31 zones and subzones established for the Australasian region and used in whole or in part in many regions around the world (Lenz and Jackson, 1986; Williams and Stevens, 1988; Bjerreskov, 1989; Ortega et al., 1993). Although the sequence of species is established from intensive sampling at many locations in Australia and New Zealand, almost none of the Australian localities have been published in the form of a measured stratigraphic section. Before the Australasian zonal range chart was incorporated in the database, Figure 7A served as an independent test of the Ordovician part of the composite sequence built from 177 sections and the 570 taxa known from at least two of these sections. After this test, the Australasian composite sequence was added to the data set as a regional zonal range chart. Eleven more regional composites were added later in this fashion.

Fourth, the scaled composite sequences may be tested against the results of manual graphic correlations that include subjective expertise. Figure 7B shows the good correspondence in the ordering of 246 graptolite range-end events between two scaled composite sequences built with and without manual assistance. The scaled Conop9 composite developed before incorporation of the Australian ranges compares favorably with Cooper's (1999) graphic composite section for the Ordovician. The late Ordovician portion of Cooper's scale is based on 61 shallow-water, carbonate conodont–bearing sections (Sweet, 1984, 1988) that are not a part of our data.

Fifth, because the scaled Conop9 composite includes dated events, it may be plotted against the standard geochronometric scale in Ma (Fig. 5). The plot was already near rectilinear ($r^2 = 0.9774$) with only 177 sections composited and before inclusion of the regional zonal range charts. Good correlation coefficients indicate that the scaled composite sequence is a nearly linear proxy for a relative time scale, *at the scale of the time intervals between dated events.*

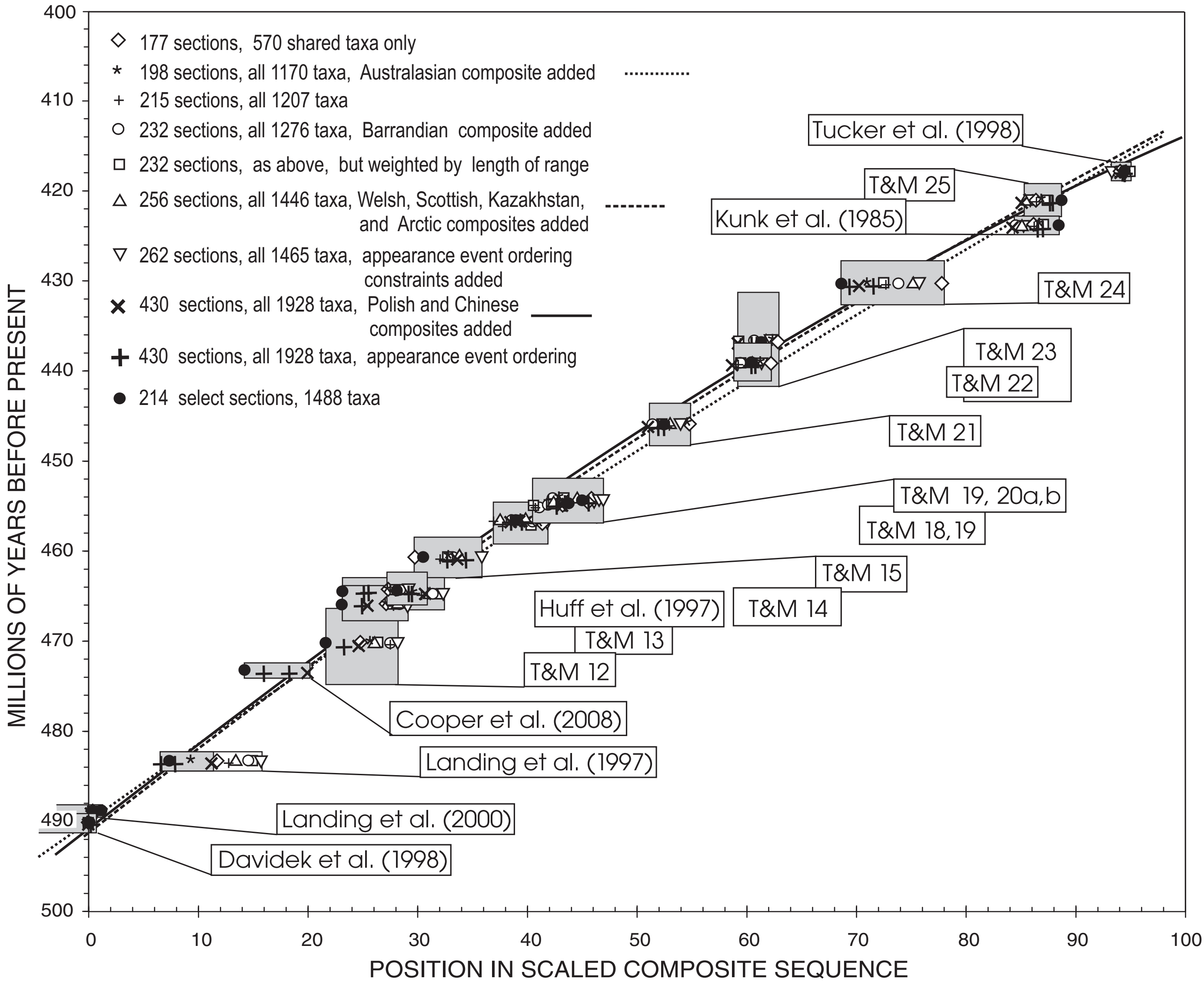


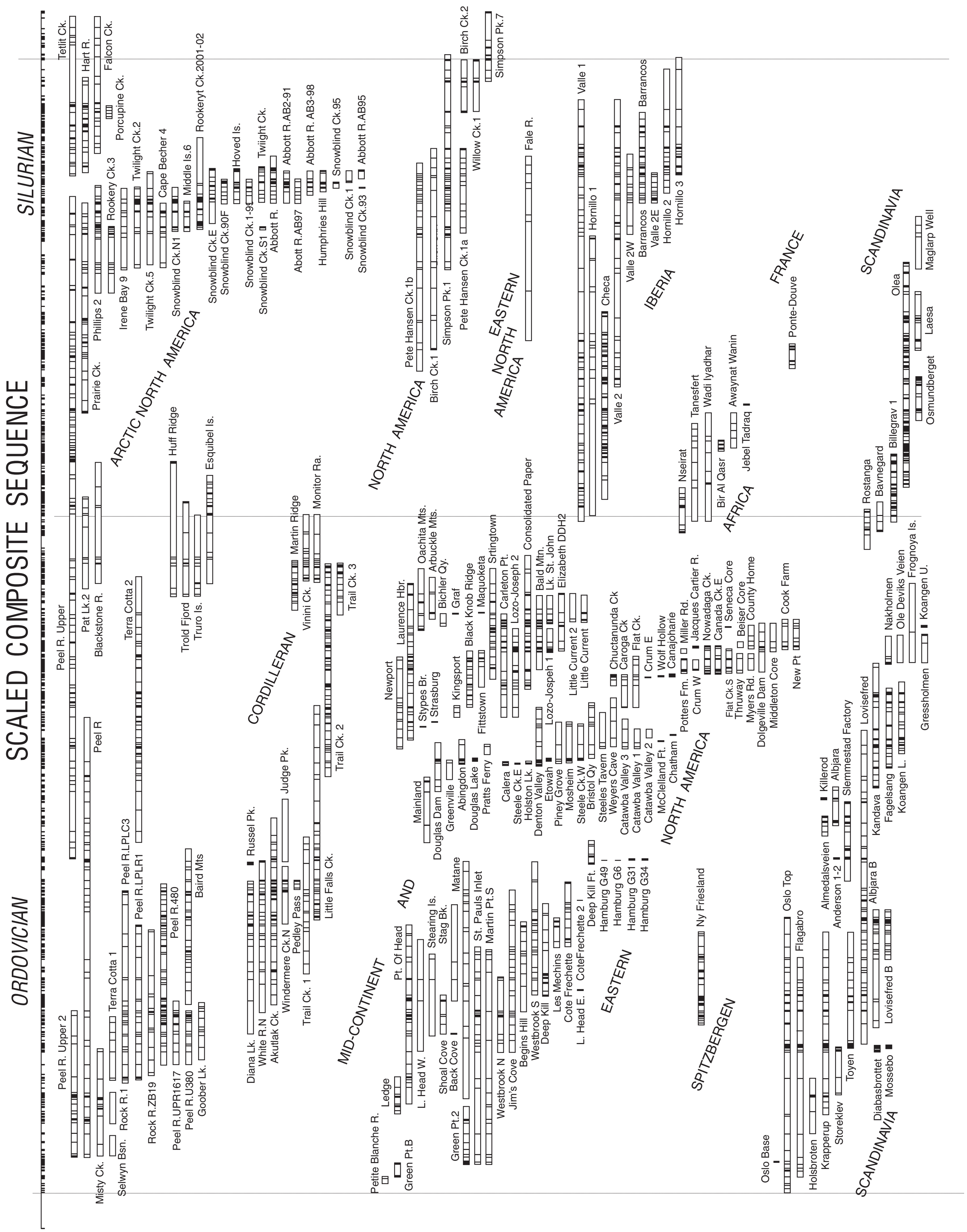
Figure 5. CONOP proxy time scales plotted against the chronometric scale for ten solutions that differ in the size of the data set and the optimization criteria. Solutions minimized range adjustments, except where appearance event ordination is stated. Rectangular boxes represent chronometric (two standard deviations) and stratigraphic errors for radioisotopically dated ash-fall events (identified by publication; "T&M" refers to numbered events in Tucker and McKerrow, 1995). The stratigraphic error is the range of positions occupied by the dated event in the different solutions. The right half of the error box for the Landing et al. (1997) date may be ignored because the associated taxon was very poorly constrained in the smaller data sets. The Cooper et al. (2008) date was not available until the data set reached 430 sections. Second-order polynomial regression lines are shown for three of the composites as indicated in the key to plotting symbols.

Sixth, the form of the regression is found to be little influenced by reasonable changes in optimization rules or the addition of new sections and taxa, even if those taxa are known from only a single section. There are no systematic shifts in the position of points within the uncertainty boxes in Figure 5. The linear regression remains strong (r^2 always approximates 0.98), and recourse to a second-order polynomial produces diminished returns as the data set grows. Initially this improved the r^2 value by 0.013; with the largest data set, the improvement in fit falls to only 0.007. The x^2 coefficient in the polynomials is always very small (–0.00000118 for 198 sections; –0.00000172 for 256 sections; –0.00000133 for 430 sections). The stratigraphic uncertainties in Figure 5 are more conservative than the best-fit intervals (Sadler and Cooper, 2003) that would be generated by the set of all equally well fit solutions arising from multiple reruns of any single data set.

RESOLVING POWER

Event Clustering

Although the large ordinal composite sequences place more than 3000 events in best-fit order, this potential resolving power cannot be fully realized in the scaled composite sequences. Sequencing uncertainty manifests itself in all composite sequences as clusters of events whose internal order cannot be resolved.


SCALED COMPOSITE SEQUENCE
ORDOVICIAN
SILURIAN
ARCTIC NORTH AMERICA
CORDILLERAN NORTH AMERICA
MID-CONTINENT AND EASTERN NORTH AMERICA
EASTERN NORTH AMERICA
SPITZBERGEN
SCANDINAVIA
AFRICA
IBERIA
FRANCE

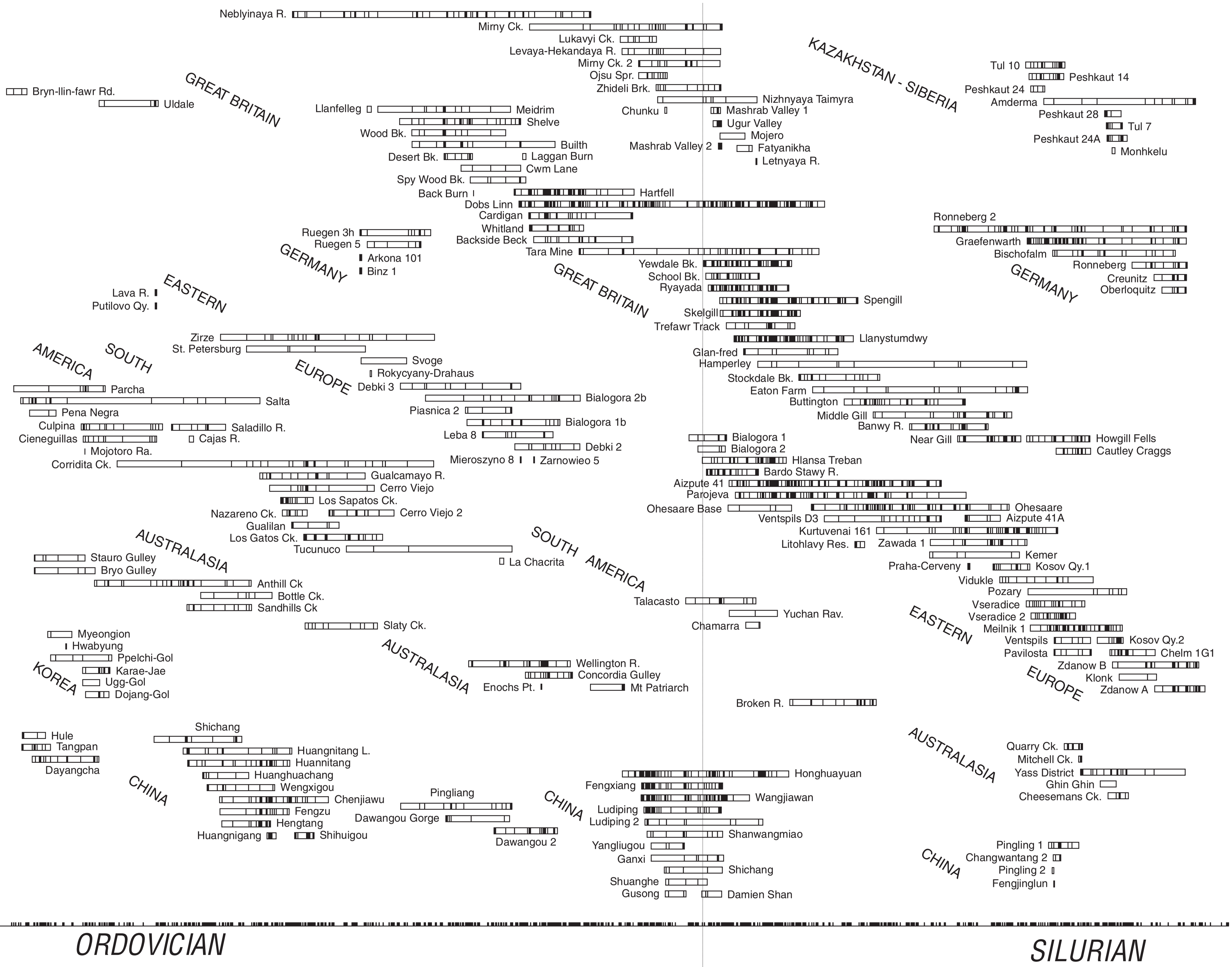


Figure 6 (*on this and previous page*). **Correlation of sections included in the optimization. The vertical scale is the scaled optimal sequence of events for the whole data set. Each local section is plotted as a rectangle that spans all paleontologic events falling within the section, after adjustment and augmentation of ranges to fit the optimal sequence. Those events actually observed in the section are represented by a black bar. Thus, the disposition of black bars represents the completeness of the global stratigraphic record of graptolites.**

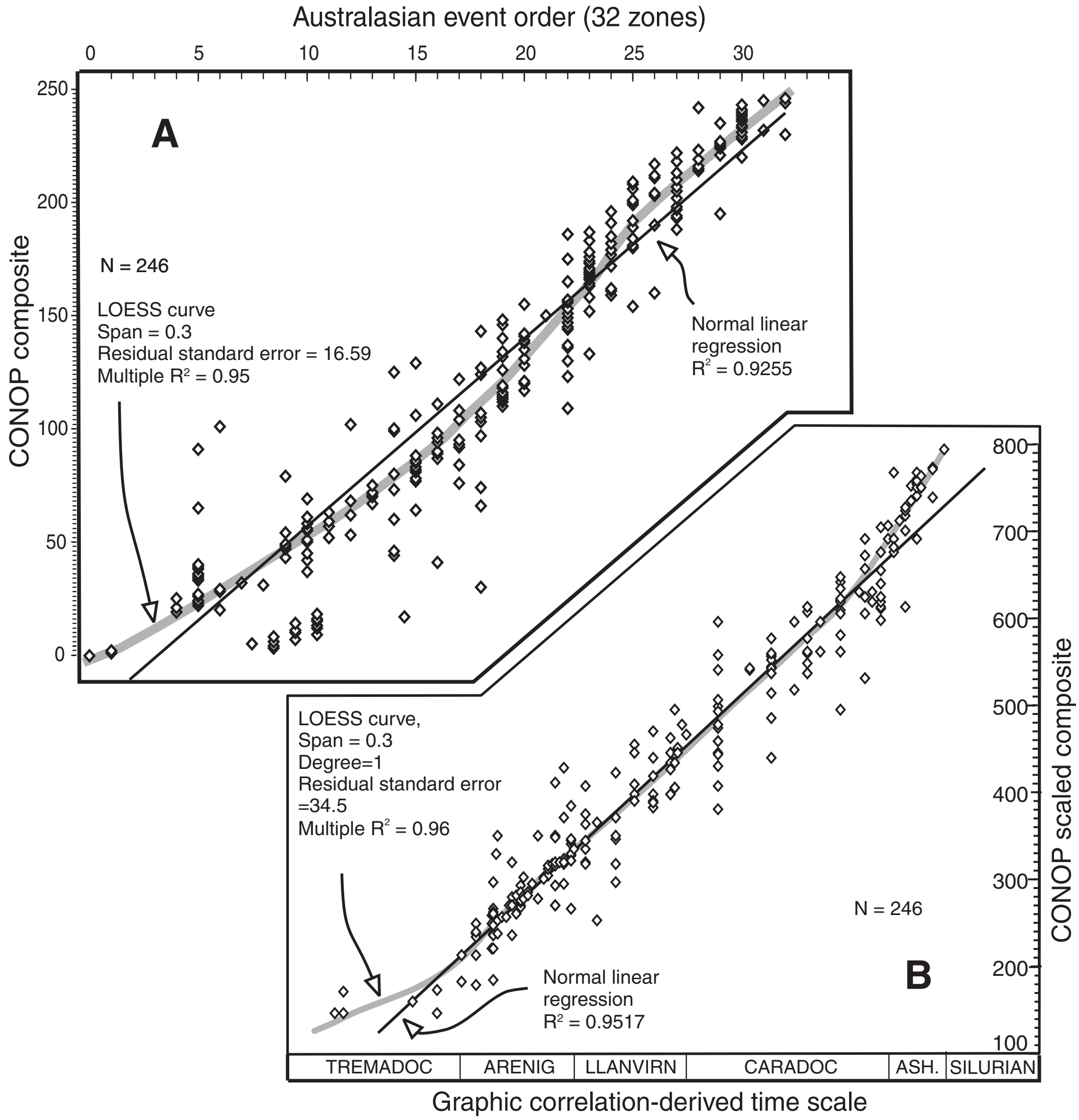


Figure 7. A—Sequence of 246 range-end events in the Australasian zonal succession plotted against their ordinal sequence in the CONOP composite as a test of the composite. B—Range-end events in the graphic correlation-based scale (Cooper, 1999) plotted against their positions in the CONOP scaled composite, as a test of the scaled composite. The data points align in vertical rows because the Australasian range chart terminates ranges only at zone or subzone boundaries. LOESS (local regression) is a computationally intensive technique that mimics nonlinear regression by fitting simpler regression models to local subsets of the data.

It is one fundamental reason why many small differences emerge between composite sequences developed from the same range charts using the same constraints and optimizing functions.

The numbers of events that cluster at a single level in the scaled composites are exponentially distributed with a mean and standard deviation of 1.9 and a maximum of 29. These values have been remarkably constant from composite 198 to composite 430. For these composites, neither events nor sections were culled in advance according to their likely resolving power; the algorithms themselves discover what a biostratigrapher would expect: not all events can be resolved. On average, then, nearly half of the events are placed at the same level as the previous event in the scaled composite. This loss in resolving power from the number of events (E) to the number of event levels (L) is linear and consistent across all scaled composite sequences (for nine scaled composite sections, L = 0.53E; r^2 = 0.89). The raw uncorrected stratigraphic sections show a similar but somewhat larger loss, having about three events on average per event level (L = 0.36E; r^2 = 0.65; n = 398). Published regional composite range charts that use only traditional zone boundaries collapse the resolving power to 12 events on average per zone (L = 0.081E; r^2 = 0.37; n = 10). Although the underdetermined nature of the problem limits the number of potentially resolvable time increments to about one-half of the number of first- and last-appearance events of graptolites, it is a significant improvement over regional zonation and a modest gain over raw stratigraphic sections. Again, the compositing process appears not to overinterpret the field information.

Resolving Power of the Scaled Composite

Using the regressions from Figure 5, the sizes of the time intervals between successive event levels in the scaled composites may be estimated in millions of years. Figure 8 summarizes the distribution of the size of these intervals through time for the most fully populated composite (composite 430); it is a guide to the potential resolving power of the current data set. In order to reveal expected (average) resolving power and general trends, the very variable values are summarized by moving-average curves. The limiting resolving power ranges between 0.1 and 0.02 m.y. and generally is finer in the Silurian than the Ordovician.

Like the sizes of irresolvable clusters, the time intervals between successive events or clusters are strongly skewed, with a preponderance of very short intervals. In composite 430, the average interval is 0.04 m.y., and the maximum is 1.38 m.y. In the selective composite 214, the average is 0.06 m.y., and the maximum is 2.14 m.y. These values calibrate the potential resolving power that remains after accounting for uncertainties in the sequencing and scaling tasks. Whether the same resolution can be realized for the calibration of graptolite zone durations depends upon the precision with which traditional zone boundaries can be located in the composite sequences. This is an issue of biostratigraphic uncertainty, discussed below.

How great a demand is placed on the scaled composite by the goal of resolving the duration of graptolite zones that might be as brief as 0.5 Ma (Zalasiewicz, 1990)? The total time span of our graptolite case study is ~77 m.y. Given that the successive composite sequences resolve from at least 1300 to more than 1900 levels, an average of 9–13 successive levels in the scaled composites would span 0.5 m.y. Thus, the scaled composite appears to have ten times the mean resolving power necessary to resolve the finest graptolite zone. For the unscaled composite sequence, this means that events need only be within ten positions of their correct order. Because the mean potential resolving power is better than 10^{-1} m.y., we interpolate ages to the nearest 10^{-2} m.y. and let the reader round up.

Time Scales of the Scaling Assumptions

Knowing the number of independently resolvable event levels, we may now estimate the average time scales of the assumptions about rates of faunal turnover and rock accumulation. The mean time span of the local stratigraphic sections, after adjustment to match the composite sequence of events, is 5.5 m.y. This is effectively the scale at which faunal turnover in the graptolite clade is assumed to be steady for the purpose of rescaling entire sections. The much shorter intervals between events and event clusters average ~100 per section or 55,000 yr each. The average number of stratigraphic sections that span each interval in the composite sequence ranges from 18 sections in composite 198 to 33 sections in composite 430. The number rarely falls below 10, except toward the ends of the study interval (Cooper and Sadler, 2004; Fig. 12.4 therein).

Scope for Improved Resolving Power

One effective way to increase practical resolving power would be to determine more radioisotopic dates for beds within sections that yield detailed graptolite range charts. Such dates strengthen the ties between different faunal provinces and further constrain the optimal sequence of range-end events. Within any one province, better resolving power would be achieved by increasing the ratio of the numbers of event levels to events—i.e., seeking more fossiliferous horizons. Both strategies aim to increase the precision of the scaled composite sequence and are limited by the preserved stratigraphic information.

Some improvement in resolving power could likely be purchased through more uniform taxonomic practices or subjective culling from the set of observed taxon ranges. The goal would be to reduce the length of composite taxon ranges and the uncertainties in recognizing zone boundaries. The composite sequences all include some highly unexpected event placements and intervals in which taxa from two to three zones are intermingled. Experts recognize these anomalies quickly, and such skills might be translatable into algorithms for quality control. Nevertheless it is a remarkable testimony to the stability of biostratigraphic practice that high resolving power emerges from a total-data approach using simple, objective, sequencing rules implemented by computer algorithms.

AN ORDOVICIAN-SILURIAN TIME SCALE

The end product of locating the zone and stage boundaries in the scaled composite and interpolating them into the chronometric scale is a calibrated time scale. Figures 9 and 10 use our preferred calibration based on composite 256 for the Silurian and the culled composite 214 for the Ordovician. These two partial composites were chosen subjectively because they accord best with the expected stratigraphic distribution of those species appearance events used in traditional zonation. However, an acceptably precise time scale could be based on any one of the four calibrated composites in Figure 5 (all calibrated values are available in Tables 1A and 1B of the supplementary materials; see footnote 1).

These estimates of the ages of stage boundaries may be compared with those of recent time scales for the Ordovician and Silurian (Fig. 11). Our scale compares most closely with that of Tucker and McKerrow (1995), particularly in the Silurian. Notice that both assign a relatively long interval to the Llandovery. This is not simply a matter of using the same dated events; our relative durations emerged from the scaling process, before calibration; they are evident in the relative time scale. We differ principally in assigning younger ages for the early Ordovician stage boundaries and in the shorter time allotted to the Llanvirn (= Llanvirn plus Llandeilo of Tucker and McKerrow). Our age assignment of 490.88 Ma for base of the Ordovician is close to that (490 Ma) of Young and Laurie (1996); but our ages for base of the Silurian and base of the Devonian are considerably older, and the proportionate durations of most of the Ordovician and Silurian stages differ markedly from their scale. Figures 9 and 10 differ somewhat from the scales published in Gradstein et al. (2004) because the data set has been substantially enlarged since then.

Figure 12 summarizes estimates of graptolite zone boundary ages for our four calibrated composite time scales. The band of estimates in Figure 12 is steepest where zone duration is briefest and horizontally widest where uncertainty is greatest, as measured by differences between the four estimates. Duration and uncertainty vary appreciably and independently. All our calibrated sequences predict that zone durations range from less than 0.1 m.y. to nearly 5.0 m.y. in the Ordovician and from less than 0.1 m.y. to more than 2 m.y. in the Silurian. None supports the assumption, once used to construct time scales, that graptolite zones are uniform in duration.

The 33 graptolite zones or groups of zones identified in the Ordovician have an average duration of 1.44 (σ = 1.25) m.y. The 31 zones and combined zones of the Silurian have a shorter average duration: 0.91 (σ = 0.62) m.y. In the most finely zoned part of the Ordovician, the Bendigonian to Yapeenian interval of 10.91 m.y., there are 11 zones, with a mean duration of 0.99 m.y. The 7.21 m.y. Silurian interval spanning the Wenlock and Ludlow contains 13 zones or zonal groups giving a mean duration of only 0.55 m.y.

UNCERTAINTY ESTIMATION

Uncertainty in time scales can be attributed to two main sources—radioisotopic

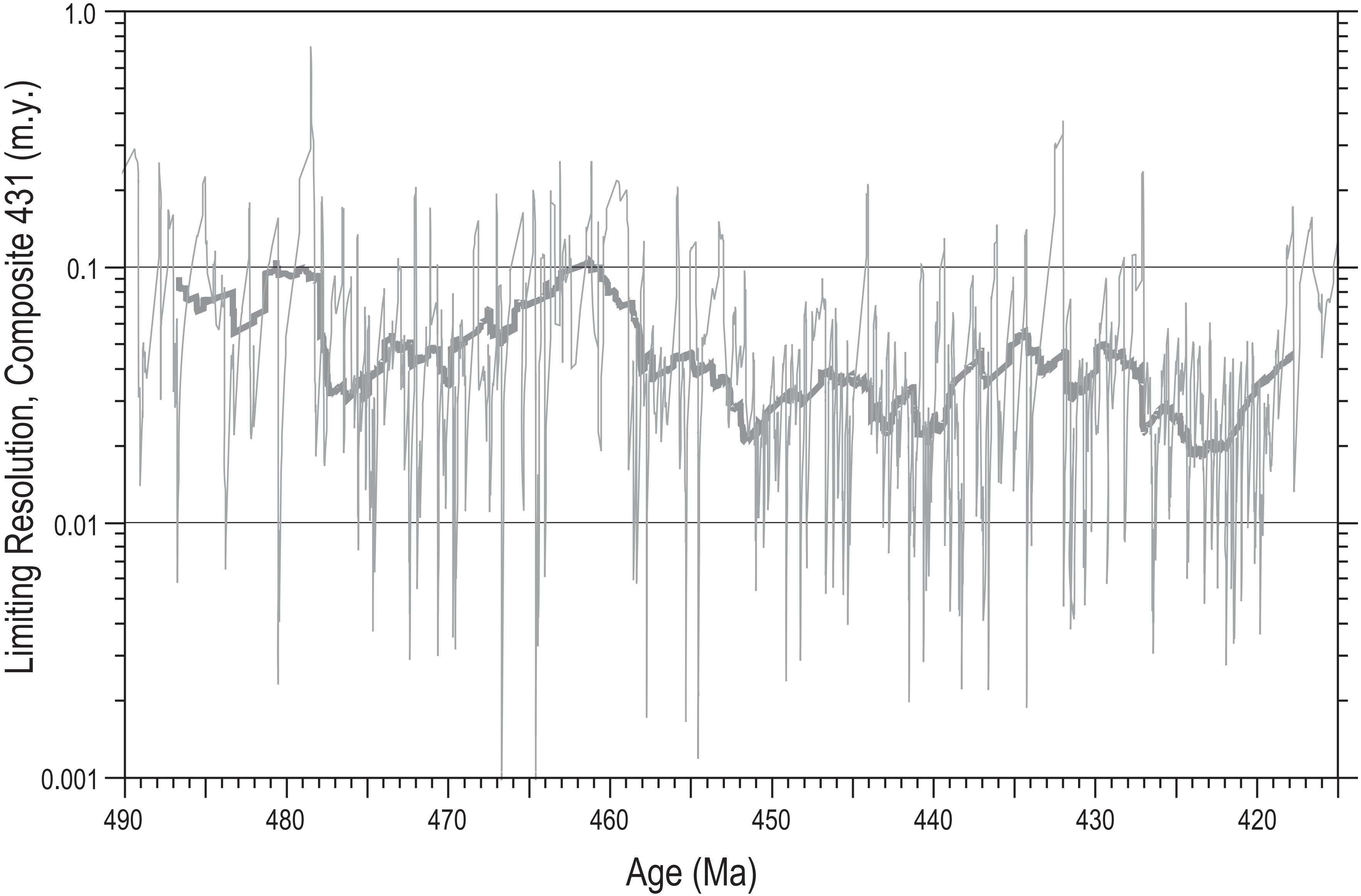


Figure 8. Variation of potential resolving power in composite 430, as a function of age. Differences in interpolated age were determined for all 3904 pairs of adjacent events. These highly variable data are summarized by 10-point (thin gray line) and 200-point moving averages (thick gray line), plotted at the center point of the moving window. All events within an irresolvable cluster are placed at the same level, and the age difference to the next higher level is used for all—i.e., irresolvable pairs of events do not generate zero values.

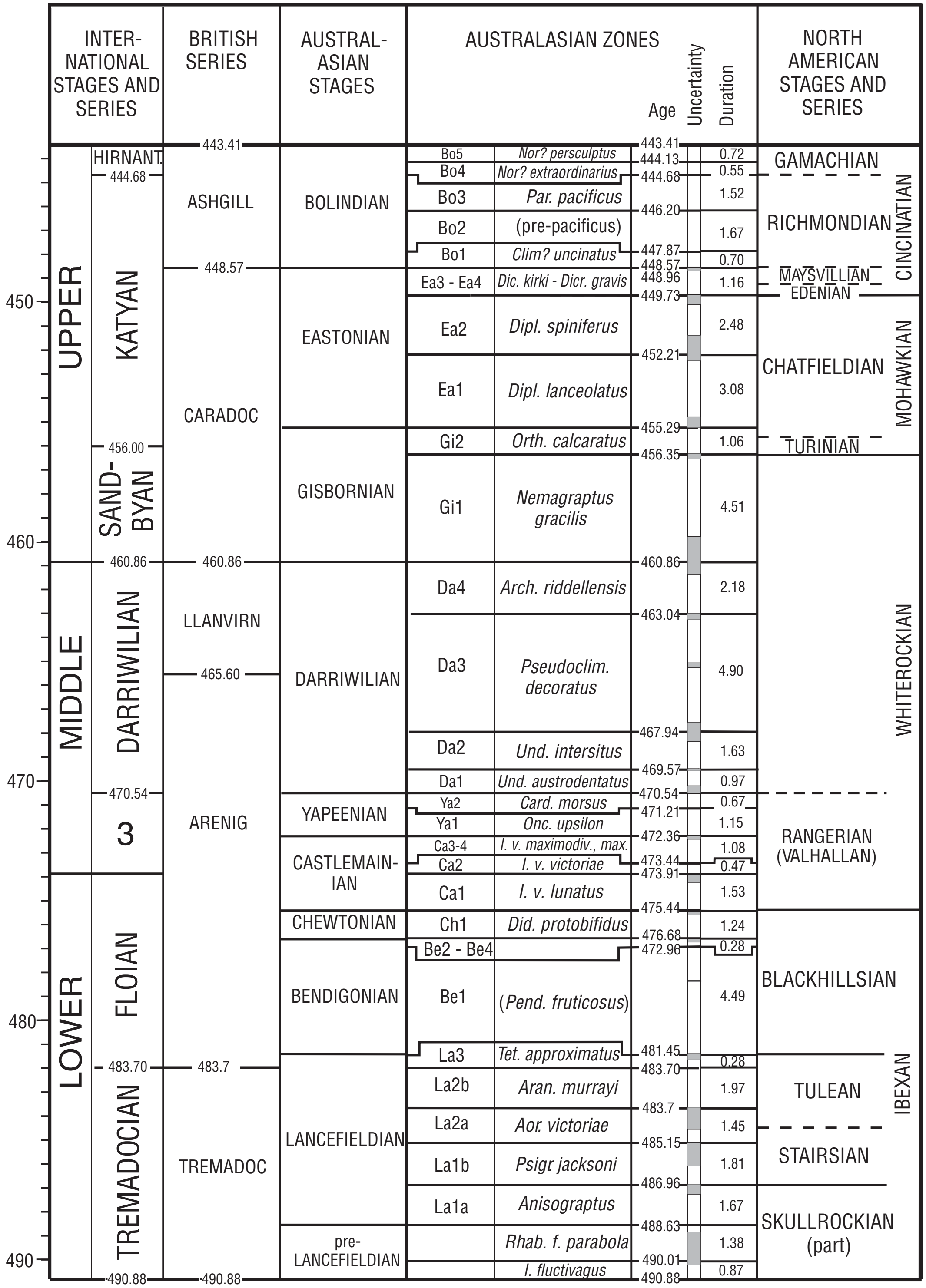


Figure 9. Preferred Ordovician time scale. The stratigraphic uncertainty indicated by gray rectangles is that arising from uncertainty in zone boundary placement. Numerical values for this and four other Ordovician-Silurian calibrations are given in Tables 1A and 1B in the GSA Data Repository (see footnote 1). Nor.—*Normalograptus;* Par.—*Paraorthograptus;* Clim.—*Climacograptus;* Dic.—*Dicellograptus;* Dicr.—*Dicranograptus;* Dipl.—*Diplacanthograptus;* Orth.—*Orthograptus;* Arch.—*Archiclimacograptus;* Pseudoclim.—*Pseudoclimacograptus;* Und.—*Undulograptus;* C.—*Cardiograptus;* Onc.—*Oncograptus;* I. v.—*Isograptus victoriae;* I. v. maximodiv, max.—*Isograptus victoriae maximodivergens; Isograptus victoriae maximus;* Did.—*Didymograptus;* Pend.—*Pendeograptus;* Tet.—*Tetragraptus;* Aran.—*Araneograptus;* Aor.—*Aorograptus;* Psigr.—*Psigraptus;* Rhab. f.—*Rhabdinopora flabeliformis;* Iap.—*Iapetognathus* (conodont).

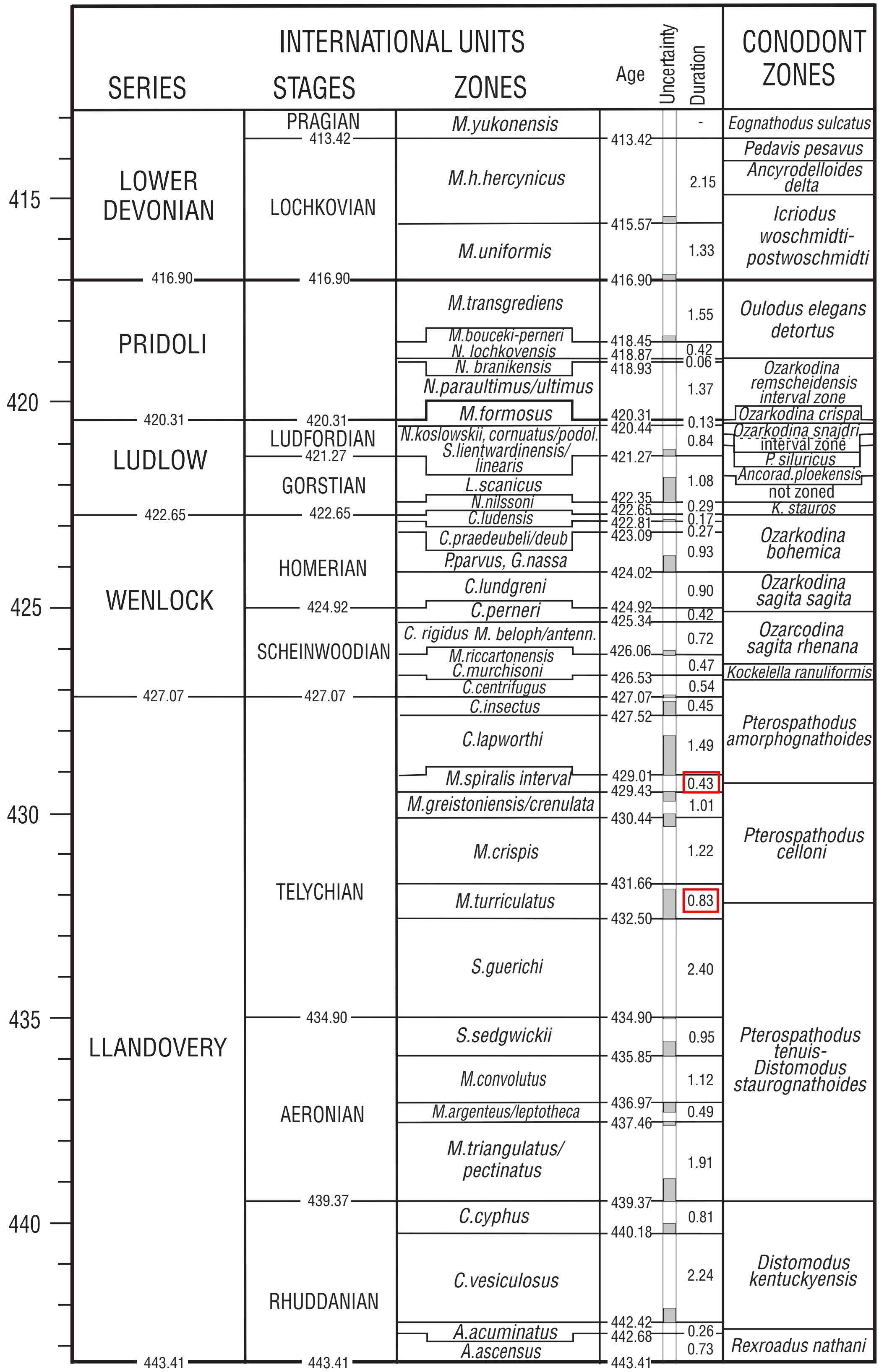


Figure 10. Preferred Silurian time scale. The stratigraphic uncertainty indicated is that arising from uncertainty in zone boundary placement. Zones with species names separated by a slash are multiple-name zones; where two or more zones are grouped, names are separated by a comma.

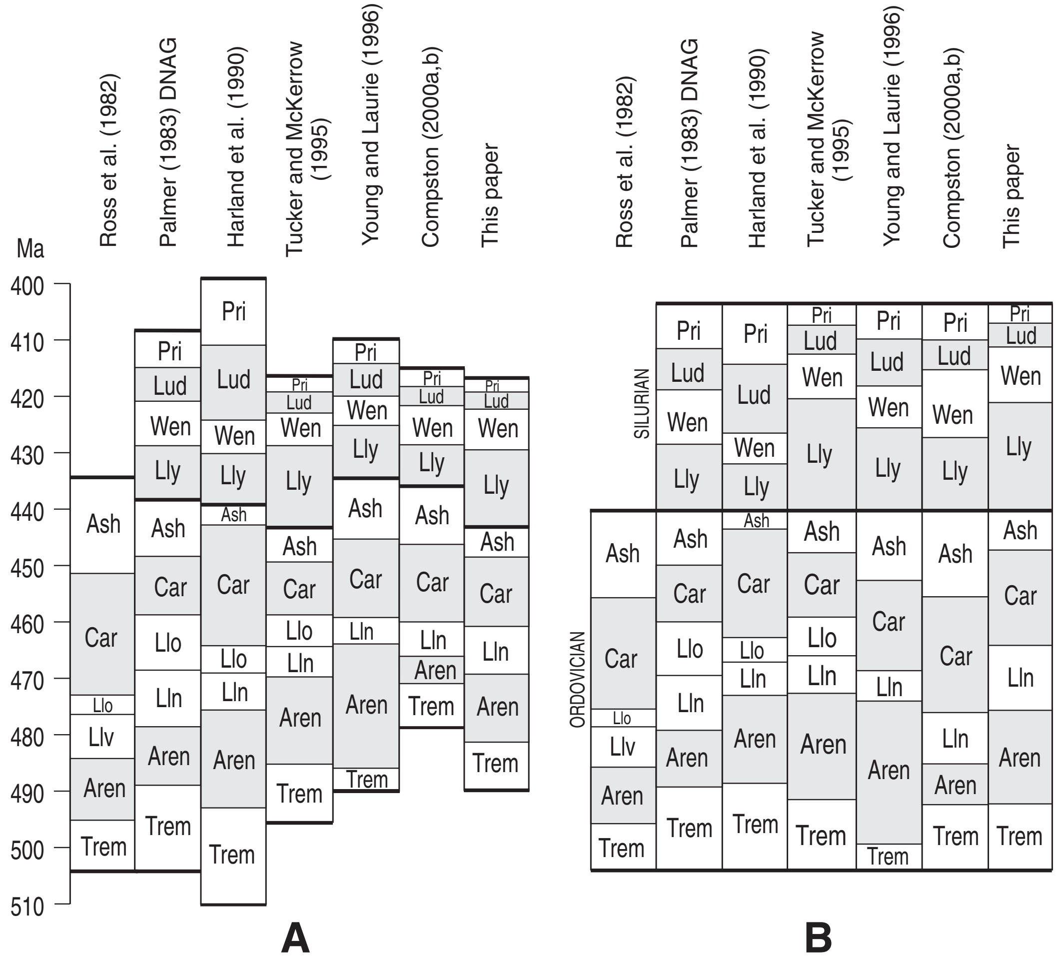


Figure 11. Comparison of the time scales for the Ordovician and Silurian Periods proposed here with those previously proposed. Ages assigned to stage boundaries are compared at left, relative proportions of stages are compared at right. The scales in GTS 2004 (Gradstein et al., 2004) are not included because they closely resemble those presented here and were built by the same technique.

uncertainty (sometimes called calibration uncertainty) and stratigraphic uncertainty. They are often depicted as the vertical and horizontal dimensions of an error box in age-by-stratigraphic position plots (Fig. 5). The treatment of this combination of uncertainties has been discussed by Agterberg (2004). The interpolation of numerical ages carries uncertainties from both radioisotopic and stratigraphic sources. The major source of uncertainty in our time scales arises from stratigraphic uncertainty, including optimization uncertainty and boundary-placement uncertainty.

Radioisotopic Uncertainty

The durations of zones, derived via the fitted regression (Fig. 5), include a stratigraphic uncertainty but do not inherit the analytical errors in the dated events, normally quoted by geochronologists as $\pm 2\sigma$ numerical error. Because we use all dated events in an ordered ensemble, the individual analytical uncertainties are much larger than the uncertainty in the calibrated ages determined by regression. Uncertainties in radioisotopic age determination are therefore not a primary limiting factor for resolving power in our time scales.

Optimization Uncertainty

As discussed above, there is some variance in the order and spacing of events within the scaled composite in repeat runs, even for the same data sets and run parameters. This is because, for most large data sets, there is no unique “best” solution but, rather, a set of equally good solutions. Essentially irresolvable clusters of events appear in different order from run to run. In smaller data sets this uncertainty can be quantified for each event, as a *best-fit interval* (Sadler and Cooper, 2003)—the interval of variance in position within successive composite sequences. In large data sets best-fit intervals become impractical to compute. Instead, this uncertainty is built into zone boundary-placement uncertainty and between-composite uncertainty.

Boundary-Placement Uncertainty

Stratigraphic uncertainty in our time scale is due largely to ambiguity in locating zone boundaries in the composite. It is measured by the range of positions in the composite sequence within which the boundary might equally well be placed. The age regressions (Fig. 5) can convert this range of uncertainty to a time interval. The exercise is subjective to the extent that a distinction has sometimes to be made between taxa that clearly are out of place because they were poorly constrained during the compositing procedure, or have been displaced due to misidentification in one of the sections, on the one hand, and those that are somewhat away from their expected level but still lie within a plausible range of natural variation (e.g., a “genuine” stratigraphically low occurrence of a FAD explicable by provinciality and migration). Events of the former category lie outside the range of boundary error. Those of the latter category deviate less and commonly are clustered in the composite. They lie within the range of

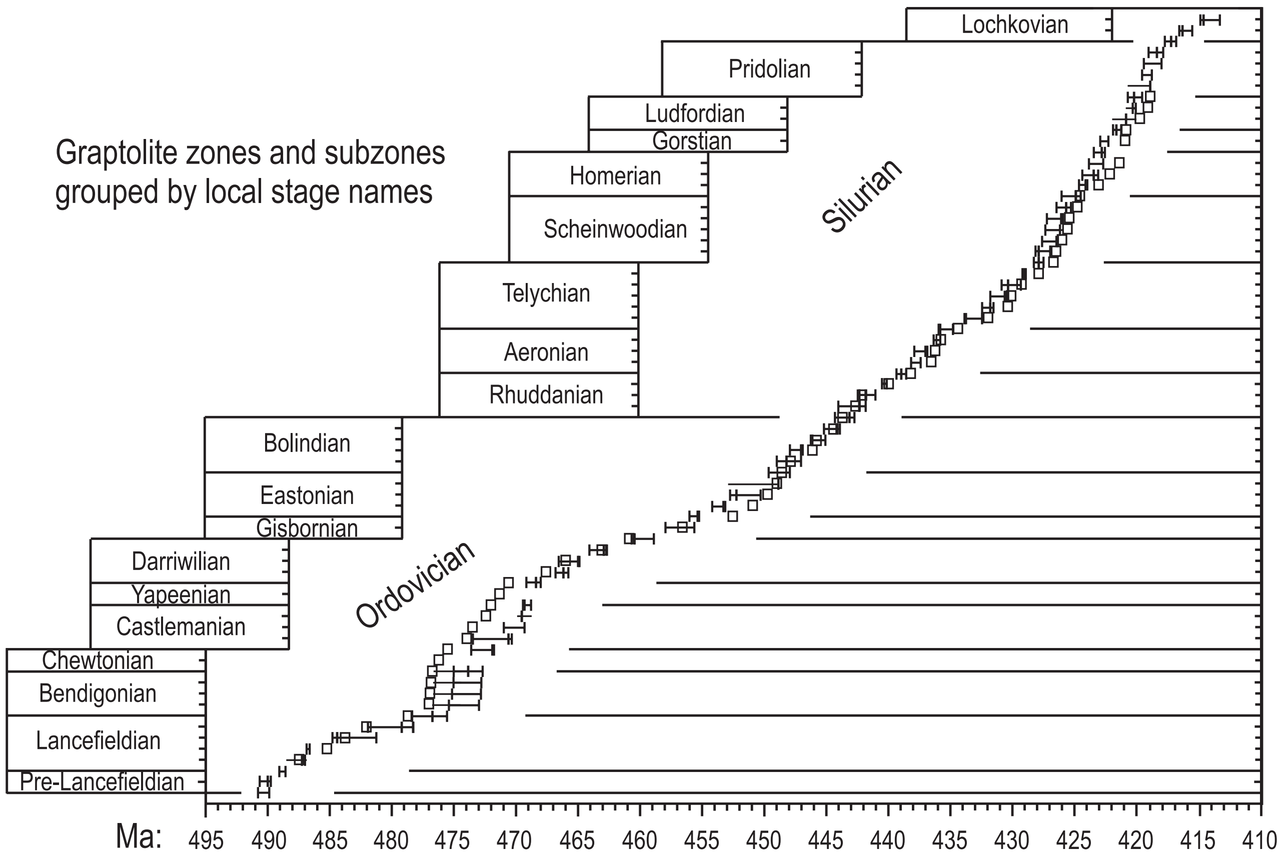


Figure 12. Between time-scale uncertainties on the age of the bases of graptolite zones as indicated by four different calibrated composite sequences. Horizontal lines with vertical bars: composites 198, 256, and 430; lines that terminate without a vertical bar indicate that the extreme value occurred in a section where the zone could not be resolved and the only constraint on the age of the base derives from the next youngest zone. Open boxes—composite 214. All numerical values in online supplementary materials, Tables 1A and 1B (see footnote 1).

boundary error. It is this stratigraphic error that is shown on our time scales (Figs. 9 and 10).

For our preferred composite sequence, the average uncertainty amounts to 0.28 (standard deviation: $\sigma = 0.37$) m.y. for the age of zone boundaries and 0.28 ($\sigma = 0.35$) m.y. for zone duration. The mean uncertainty on the ages of Ordovician zone boundaries is 0.36 ($\sigma = 0.46$) m.y., and on Silurian zone boundaries is 0.22 ($\sigma = 0.26$) m.y. In relative terms, the average *within*-composite uncertainty on the *duration* of a zone is 0.28 ($\sigma = 0.29$) of the zone duration in the Ordovician and 0.24 ($\sigma = 0.25$) of the Silurian zone durations.

Between-Composite Uncertainty

The most conservative estimates of uncertainty derive from the comparison of the ten composite sequences based upon different data or optimization rules (Figs. 5 and 12). The differences between these sequences combine two effects: the lack of a unique solution to any one data set plus the genuine improvements that result from adding more and better data. It is not possible to tease these apart completely. The average difference in age estimates resulting from a change in composite is close to the expected (mean) duration of zones for both the Ordovician and Silurian: 1.50 m.y. for the Ordovician and 0.87 m.y. for the Silurian. The largest age difference arises in an initially troublesome time interval where additions to the data set reduced the stratigraphic uncertainty and improved the results. Where the changes appear unsystematic, yielding no obvious advantages, the mean uncertainty is smaller (0.95 m.y. as compared with 1.17 m.y. overall). This suggests that the computer-assisted analysis has not overinterpreted the field data. Graptolite zone boundaries usually represent the practical limits of reproducible and widely applicable correlation by subjective biostratigraphic expertise without computer assistance. Faced with different optimization criteria, the computer algorithms find a similar limit.

These uncertainties arise from different data sets and/or optimization rules. Although the different objective functions may be true alternatives, the stratigraphic information must generally be expected to improve as the data set grows—especially via inclusion of the most recent range charts and taxonomic revisions. Thus, these uncertainties should be very conservative estimates of stability, and it is not appropriate to attach them to a time scale based on a single composite and data set. The uncertainties in Figures 9 and 10 reflect the size of biostratigraphic uncertainties that are applicable to single calibrated composite sequences.

The estimated durations of zones are less sensitive to these changes, differing by 0.44 m.y. ($\sigma = 0.34$) on average for Silurian zones and 0.93 m.y. ($\sigma = 0.84$) for Ordovician zones across the four calibrated composites, and averaging 0.68 m.y. overall. The average graptolite zone duration is 1.17 m.y. In the Ordovician, then, graptolite zones are of longer duration on average and subject to greater uncertainty than in the Silurian.

This mirrors the differences in potential resolving power of the composite time line in the two periods. The Ordovician interval in which zone boundary age is least certain (late Lancefieldian to Yapeenian, Fig. 12), although finely zoned, coincides with a time of marked provincialism in the distribution of graptolite faunas (Bulman, 1971). It is also in this interval that the successive composites produced a systematic improvement in the sequencing of index taxa.

SUMMARY AND CONCLUSIONS

(1) Conop9 optimization algorithms (Sadler et al., 2003) may be used to order over 3800 graptolite range-end events from more than 400 sections worldwide into a single global ordinal composite sequence that best fits the observed order of events in the local sections. Radioisotopically dated events can be allowed to find their natural position among the ordered paleontologic events, according to the fossil taxa associated with the dated bed.

(2) The optimization process stacks the local sections, all of which are substantially shorter than the total span of the time scale, by combining two measures of the misfit between model sequences and the sections—the need to extend observed ranges and to imply additional coexistences.

(3) The optimal composite sequence of events can then be scaled in time using two traditional simplifications, but at different time scales. The net amount of biological change (number of species appearance and disappearance events) may approximate the relative duration of long time intervals—the time span of whole stratigraphic sections. Relative duration of the much shorter time intervals between successive event levels may be estimated from rock thicknesses, averaged across many stratigraphic sections, *after* the range charts have been "normalized" to reduce the effects of variance in depositional rate.

(4) The radiometrically dated levels embedded in the composite show the near linearity of the scaled composite and demonstrate the feasibility of the optimizing and scaling procedures.

(5) Zone and stage boundaries can be located in the scaled composite, converting it to a relative geologic time scale. Not all zonal index fossils appear in the composite in the same order as in the local zonal schemes. Many zones must be recognized by a cluster of characteristic species; thus local interval zones may become assemblage zones in the global context.

(6) The radiometrically dated levels support calibration of the relative scale by interpolation. The resulting calibrated time scale (geochronologic scale) for the Ordovician and Silurian improves the resolution over the best resolved parts of previous zone-based scales by about ten times. Mean duration of an Ordovician graptolite zone is 1.44 ($\sigma = 1.25$) m.y. and of a Silurian zone is 0.91 ($\sigma = 0.62$ m.y.).

(7) The main source of uncertainty within the final scale is stratigraphic uncertainty arising from uncertain placement of stage and zone boundaries in the scaled composite sequence. This uncertainty averages 0.28 m.y. for the age of zone boundaries and for zone duration.

ACKNOWLEDGMENTS

We thank C. Mitchell for assistance in locating the Late Ordovician and early Silurian zonal boundaries in the CONOP composite. R. Fortey, D. Goldman, J. Maletz, C. Mitchell, F. Vandenberg, and J. Zalasiewicz provided unpublished biostratigraphic data. S. Bowring, W. Compston, and M. Villeneuve gave valuable comments and opinions on the age interpretation of Ordovician and Silurian zircon-dated volcanics. F. Agterberg, H. Peters, and D. Rhoades provided advice and assistance with curve fitting. M. Cooper, Q. Crowley, and A. Rushton kindly made details of their zircon date on the Tyrone Volcanic Group available to us in advance of publication. The constructive comments from B. Pratt, R. Scott, and B. Kauffman on an early version of the paper resulted in a considerably clearer presentation. Development of the Conop9 program was partially supported by National Science Foundation grants EAR 9980372, EAR 0518939, and OPP 0338274 to Sadler.

REFERENCES CITED

Agterberg, F.P., 2004, Appendix 3—Geomathematics, *in* Gradstein, F., Ogg, J.G., and Smith, A.G., eds., A geologic time scale 2004: Cambridge, Cambridge University Press, p. 485–486.

Alroy, J., 1992, Conjunction among taxonomic distributions and the Miocene mammalian biochronology of the Great Plains: Paleobiology, v. 18, p. 326–343.

Alroy, J., 1994, Appearance event ordination: A new biochronologic method: Paleobiology, v. 20, p. 191–207.

Armstrong, H.A., 1995, High-resolution biostratigraphy (conodonts and graptolites) of the Upper Ordovician and Lower Silurian—A chronometric framework in which to evaluate the causes of mass extinction: Modern Geology, v. 20, p. 41–68.

Bellmore, M., and Nemhauser, G.L., 1968, The traveling salesman problem: A survey: Operations Research, v. 16, p. 538–558.

Berggren, W.A., Hilgen, F.J., Langereis, C.G., Kent, D.V., Obradovich, J.D., Raffi, I., Raymo, M.E., and Shackleton, N.J., 1995, Late Neogene chronology: New perspectives in high-resolution stratigraphy: Geological Society of America Bulletin, v. 107, p. 1272–1287.

Bergström, S.M., 1986, Biostratigraphic integration of Ordovician graptolite and conodont zones—A regional review, *in* Rickards, R.B., and Hughes, C.P., eds., Paleobiology and biostratigraphy of graptolites: Oxford, Blackwell Scientific, p. 61–78.

Bergström, S.M., and Mitchell, C.E., 1992, The Ordovician Utica Shale in the eastern Midcontinent region: Age, lithofacies, and regional relationships: Oklahoma Geological Survey Bulletin, v. 145, p. 67–89.

Bergström, S.M., and Wang, Z.-H., 1995, Global correlation of Castlemainian to Darriwilian conodont faunas and their relations to the graptolite zone succession, *in* Xu, C., and Bergström, S.M., eds., The base of the Austrodentatus Zone as a level for global subdivision of the Ordovician System: Nanjing, Nanjing University Press, p. 92–98.

Berry, W.B.N., 1979, Graptolite biogeography: A biogeography of some Lower Paleozoic Plankton, *in* Gray, J., and Boucot, A.J., eds., Historical biogeography, plate tectonics and the changing environment: Corvallis, Proceedings of the Annual Biology Colloquium and Selected Papers, Oregon State University Press, p. 105–115.

Bjerreskov, M., 1989, Ordovician graptolite biostratigraphy in North Greenland: Rapport Gronlands Geologiske Undersogelse, v. 144, p. 17–33.

Boucot, A.J., 1975, Evolution and extinction rate controls: Amsterdam, Elsevier, 427 p.

Bowring, S.A., Grotzinger, J.P., Isachsen, C.E., Knoll, A.H., Pelechaty, S.M., and Kolosov, P., 1993, Calibrating rates of Early Cambrian evolution: Science, v. 261, p. 1293–1298, doi: 10.1126/science.11539488.

Bulman, O.M.B., 1971, Graptolite faunal distribution, *in* Middlemiss, F.A., Rawson, P.F., and Newall, G., eds., Faunal provinces in space and time: Special Issue of the Geological Journal, p. 47–60.

Carter, C., Trexler, J.H., and Churkin, M., 1980, Dating of graptolite zones by sedimentation rate: Implications for rates of evolution: Lethaia, v. 13, p. 279–287, doi: 10.1111/j.1502-3931.1980.tb01051.x.

Chen, X., Zhang, Y., and Mitchell, C.E., 2001, Early Darriwilian graptolites from central and western China: Alcheringa: An Australasian Journal of Palaeontology, v. 25, p. 191–210.

Churkin, M., Carter, C., and Johnson, B.R., 1977, Subdivision of Ordovician and Silurian time scale using accumulation rates of graptolitic shale: Geology, v. 5, p. 452–456, doi: 10.1130/0091-7613(1977)5<452:SOOAST>2.0.CO;2.

Cocks, L.R.M., 1971, Facies relationships in the European Lower Silurian: Mémoire du Bureau de Recherches Géologique et Minières, v. 73, p. 223–227.

Cocks, L.R.M., and Torsvik, T.H., 2002, Earth geography from 500 to 400 million years ago: A faunal and palaeomagnetic review: Journal of the Geological Society, v. 159, p. 631–644, doi: 10.1144/0016-764901-118.

Compston, W., 2000a, Interpretations of SHRIMP and isotope dilution zircon ages for the Paleozoic time-scale: I. The early Ordovician and late Cambrian: Mineralogical Magazine, v. 64, p. 43–57.

Compston, W., 2000b, Interpretations of SHRIMP and isotope dilution zircon ages for the Paleozoic time-scale: II. Silurian to Devonian. Mineralogical Magazine v. 64, p. 1127–1146.

Cooper, M.R., Crowley, Q.G., and Rushton, A.W.A., 2008, New age constraints for the Ordovician Tyrone Volcanic Group, Northern Ireland: Journal of the Geological Society, v. 165, p. 333–339, doi: 10.1144/0016-76492007-057.

Cooper, R.A., 1992, A relative time scale for the Early Ordovician derived from depositional rates of graptolitic shales, *in* Webby, B.D., and Laurie, J.R., eds., Global perspectives on Ordovician geology: Balkema, Rotterdam, p. 3–21.

Cooper, R.A., 1998, Towards a general model for the depth ecology of graptolites, *in* Gutiérrez-Marco, J.C., and Rabano, I., eds., Proceedings of the 6th International Graptolite Conference (GWG-IPA) and 1988 field meeting: Temas Geológico-Mineros ITGE, p. 161–163.

Cooper, R.A., 1999, The Ordovician time scale—Calibration of graptolite and conodont zones, in Kraft, P., and Fatka, O., eds., Quo vadis Ordovician? Short papers of the 8th International Symposium on the Ordovician System: Acta Universitatis Carolinae, Geologica, v. 43, p. 1–4.

Cooper, R.A., and Sadler, P.M., 2004, Chapter 12, Ordovician, *in* Gradstein, F.M., Ogg, J.G., and Smith, A.G., eds., A geologic time scale 2004: Cambridge, Cambridge University Press, p. 165–187.

Cooper, R.A., Crampton, J.S., Raine, J.I., Gradstein, F.M., Morgans, H.E.G., Sadler, P.M., Strong, C.P., Waghorn, D., and Wilson, G.J., 2001, Quantitative biostratigraphy of the Taranaki Basin, New Zealand—A deterministic and probabilistic approach: American Association of Petroleum Geologists Bulletin, v. 85, p. 1469–1498.

Davidek, K., Landing, E., Westrop, S.R., Rushton, A.W.A., Fortey, R.A., and Adrain, J.M., 1998, New uppermost Cambrian U-Pb date from Avalonian Wales and the age of the Cambrian-Ordovician boundary: Geological Magazine, v. 133, p. 305–309.

Edwards, L.E., 1978, Range charts and no-space graphs: Computers & Geosciences, v. 4, p. 247–258, doi: 10.1016/0098 -3004(78)90057-2.

Finney, S.C., and Berry, W.B.N., 1997, New perspectives on graptolite distributions and their use as indicators of platform margin dynamics: Geology, v. 25, p. 919–922, doi: 10.1130/0091-7613(1997)025<0919:NPOGDA>2.3.CO;2.

Gale, N.H., 1985, Numerical calibration of the Palaeozoic time scale; Ordovician, Silurian and Devonian Periods, *in* Snelling, N.J., ed., The chronology of the geological record: The Geological Society of London Memoir, v. 10, p. 81–88.

Gradstein, F.M., Ogg, J.G., and Smith, A.G., 2004, A geologic time scale 2004: Cambridge, Cambridge University Press, 598 p.

Guex, J., 1991, Biological correlations: Berlin, Springer-Verlag, 252 p.

Harland, W.B., Armstrong, R.L., Cox, A.V., Craig, L.E., Smith, A.G., and Smith, D.G., 1990, A geologic time scale: Cambridge, Cambridge University Press, 263 p.

Holland, C.H., and Bassett, M.G., 1989, A global standard for the Silurian system: National Museum of Wales, Geological Series, v. 9, p. 265–274.

Hood, K.C., 1986, Graphcor, interactive graphic correlation for microcomputers, version 2.2: Houston, K.C. Hood.

Huff, W.D., Davis, D., Bergström, S.M., Krekeler, M.P.S., Kolata, D.R., and Cingolani, C., 1997, A biostratigraphically well-constrained K-bentonite U-Pb zircon age of the lowermost Darriwilian Stage (Middle Ordovician) from the Argentine Precordillera: Episodes, v. 20, p. 29–33.

Kemple, W.G., Sadler, P.M., and Strauss, D.J., 1989, A prototype constrained optimization solution to the time correlation problem, *in* Agterberg, F.P., and Bonham-Carter, G.F., eds., Statistical applications in the earth sciences: Geological Survey of Canada Paper: Ottawa, p. 417–425.

Kemple, W.G., Sadler, P.M., and Strauss, D.J., 1995, Extending graphic correlation to many dimensions: Stratigraphic correlation as constrained optimization, *in* Mann, K.O., and Lane, H.R., eds., Graphic correlation: Society of Economic Paleontologists and Mineralogists Special Publication, v. 53, p. 65–82.

Kirkpatrick, S., Gelatt, C.D., and Vecchi, M.P., 1983, Optimization by simulated annealing: Science, v. 220, p. 671–680, doi: 10.1126/science.220.4598.671.

Kleffner, M.A., 1989, A conodont-based Silurian chronostratigraphy: Geological Society of America Bulletin, v. 101, p. 904–912, doi: 10.1130/0016-7606(1989)101<0904:ACBSC>2.3.CO;2.

Kleffner, M.A., 1995, A conodont- and graptolite-based Silurian chronostratigraphy, *in* Mann, K.O., and Lane, H.R., eds., Graphic correlation: Society of Economic Paleontologists and Mineralogists Special Publication, v. 53, p. 159–176.

Kominz, M.A., and Bond, G.C., 1990, A new method of testing periodicity in cyclic sediments—Application to the Newark Supergroup: Earth and Planetary Science Letters, v. 98, p. 233–244, doi: 10.1016/0012-821X(90)90062-3.

Koren', T.N., Lenz, A.C., Loydell, D.K., Melchin, M.J., Storch, P., and Teller, L., 1996, Generalized graptolite zonal sequence defining Silurian time intervals for global paleogeographic studies: Lethaia, v. 29, p. 59–60.

Kunk, M.J., Sutter, J., Obradovich, J.D., and Lanphere, M.A., 1985, Age of biostatigraphic horizons within the Ordovician and Silurian systems, *in* Snelling, N.J., ed., The chronology of the geological record: Oxford, Blackwell Scientific Publications, p. 89–92.

Landing, E., Bowring, S.A., Fortey, R.A., and Davidek, K.L., 1997, U-Pb zircon date from Avalonian Cape Breton Island and geochronologic calibration of the Early Ordovician: Canadian Journal of Earth Sciences, v. 34, p. 724–730.

Landing, E., Bowring, S.A., Davidek, K.L., Rushton, A. W. A., Fortey, R. A., and Wimbledon, W. A. P., 2000, Cambrian-Ordovician boundary age and duration of the lowest Ordovician Tremadoc series based on U-Pb zircon dates from Avalonian Wales: Geological Magazine, v. 137, p. 485–494.

Lenz, A.C., and Jackson, D.E., 1986, Arenig and Llanvirn graptolite biostratigraphy, Canadian Cordillera, *in* Hughes, C.P., and Rickards, R.B., eds., Palaeoecology and biostratigraphy of graptolites: The Geological Society of London Special Publications, v. 20, p. 27–45.

Löfgren, A., 1996, Lower Ordovician conodonts, reworking, and biostratigraphy of the Orreholmen quarry, Vastergötland, south-central Sweden: Lethaia, v. 118, p. 169–183.

McKerrow, W.S., Lambert, R.St.J., and Chamberlain, V.E., 1980, The Ordovician, Silurian, and Devonian time scales: Earth and Planetary Science Letters, v. 51, p. 1–8, doi: 10.1016/0012-821X(80)90250-2.

McKerrow, W.S., Lambert, R.St.J., and Cocks, L.R.M., 1985, The Ordovician, Silurian, and Devonian Periods, *in* Snelling, N.J., ed., The chronology of the geological record: The Geological Society of London Memoir, v. 10, p. 73–80.

Melchin, M., Cooper, R.A., and Sadler, P.M., 2004, Chapter 13, Silurian system, *in* Gradstein, F.M., Ogg, J.G., and Smith, A.G., eds., A geologic time scale 2004: Cambridge University Press, p. 188–201.

Murphy, M.A., and Edwards, L.E., 1977, The Silurian-Devonian boundary in central Nevada, *in* Murphy, M.A., Berry, W.B.N., and Sandberg, C.A., eds., Western North American Devonian: Riverside, University of California, Campus Museum Contributions, v. 4, p. 183–189.

Odin, G.S., 1985, Some key rules for calibration of the numerical time scale, *in* Snelling, N.J., ed., The chronology of the geological record, The Geological Society of London Memoir, v. 10, p. 41–46.

Ortega, G., Toro, B., and Brussa, E., 1993, Las de Graptolitos de la Formación Gualcamayo (Arenigiano Tardío-Llanvirniano Temprano) en el Norte de la Precordillera (Provincias de la Rioja y San Juan), Argentina: Revista Espanola de Paleontologia, v. 8, p. 207–219.

Palmer, A.R., 1983, The decade of North American geology, 1983, Geologic time scale: Geology, v. 11, p. 503–504, doi: 10.1130/0091-7613(1983)11<503:TDONAG>2.0.CO;2.

Ross, R.J., Jr., Adler, F.J., Amsden, T.W., Bergstrom, D., Bergström, S. M., Carter, C., Churkin, M., Jr., Cressman, E.R., Derby, J.R., Dutro, J.T., Jr., Ethington, R.L., Finney, S.C., Fisher, D.W., Fisher, J.H., Harris, A.G., Hintze, L.F., Ketner, K.B., Kolata, D.L., Landing, E., Neumann, R.B., Pojeta, J. Jr., Potter, A.W., Rader, E.K., Repetski, R.H., Ross, J.R.P., Shaver, R.H., Sweet, W.C., Thompson, T.L., and Webers, G.F., 1982, The Ordovician System in the United States, 1982, Correlation Chart and Explanatory Notes: Paris, International Union of Geological Sciences, Publication, v. 12, 73 p.

Sadler, P.M., 2004, Quantitative biostratigraphy: Achieving finer resolution in global correlation: Annual Review of Earth and Planetary Sciences, v. 32, p. 187–213, doi: 10.1146/annurev.earth.32.101802.120428.

Sadler, P.M., and Cooper, R.A., 2003, Best-fit intervals and consensus sequences—A comparison of the resolving power of traditional biostratigraphy and computer assisted correlation, *in* Harries, P.J., ed., High resolution approaches in stratigraphic paleontology: Topics in geobiology, v. 21, Dordrecht, Kluwer Academic Publishers, p. 49–94.

Sadler, P.M., and Cooper. R.A., 2004, Calibration of the Ordovician Time Scale, *in* Webby, B., Paris, F., Droser, M.L., and. Percival, I. G., eds., The great Ordovician biodiversification event, New York, Columbia University Press, p. 48-51.

Sadler, P.M., and Kemple, W.G., 1995, Using rapid, multidimensional, graphic correlation to evaluate chronostratigraphic models for the Mid-Ordovician of the Mohawk Valley, New York, *in* Cooper, J.D., Droser, M.L., and Finney, S.F., eds., Ordovician odyssey: Short papers for the Seventh International Symposium on the Ordovician System: Fullerton, California, Society for Sedimentary Geology (SEPM), Pacific Section, v. 77, p. 257–260.

Sadler, P.M., Kemple, W.G., and Kooser, M.A., 2003, Contents of the compact disk—CONOP9 programs for solving the stratigraphic correlation and seriation problems as constrained optimization, *in* Harries, P.J., ed., High resolution approaches in stratigraphic paleontology: Dordrecht, Topics in Geobiology, v. 21, Kluwer Academic Publishers, p. 461–465.

Salvador, A., 1994, International stratigraphic guide—A guide to stratigraphic classification, terminology, and procedure: International Union of Geological Sciences and Geological Society of America, 214 p.

Savary, J., and Guex, J., 1999, Discrete biochronological scales and unitary associations: Description of the BioGraph computer program: Lausanne, Mémoires de Géologie, Section des Sciences de la Terre, University Université de Lausanne, 281 p.

Scott, G.H., 1985, Homotaxy and biostratigraphical theory: Palaeontology, v. 28, p. 777–782.

Shaw, A.B., 1964, Time in stratigraphy: New York, McGraw-Hill, 365 p.

Skevington, D., 1974, Controls influencing the composition and distribution of Ordovician graptolite faunal provinces: Special Papers in Palaeontology, v. 13, p. 59–73.

Smith, A.G., 1993, Methods for improving the chronometric time-scale, *in* Hailwood, E.A., and Kidd, R.B., eds., High resolution stratigraphy: The Geological Society of London Special Publication, v. 70, p. 9–25.

Sweet, W.C., 1984, Graphic correlation of upper Middle and Upper Ordovician rocks, North American Midcontinental Province, USA, *in* Bruton, D.L., ed., Aspects of the Ordovician system: Paleontological Contributions of the University of Oslo, v. 295, p. 23–35.

Sweet, W.C., 1988, Mohawkian and Cincinnatian chronostratigraphy: Bulletin, New York State Museum, v. 462, p. 84–90.

Sweet, W.C., 1995, Graphic assembly of a conodont-based composite standard for the Ordovician system of North America, *in* Mann, K.O., and Lane, H.R., eds., Graphic correlation: Society of Economic Paleontologists and Mineralogists Special Publication, v. 53, p. 139–150.

Sweet, W.C., and Tolbert, C.M., 1997, An Ibexian (Lower Ordovician) reference section in the Southern Egan Range, Nevada, for a conodont-based chronostratigraphy: U.S. Geological Survey Professional Paper 1579, p. 53–84.

Sweet, W.C., Ethington, R.L., and Harris, A.G., 2005, A conodont-based reference section in central Nevada for the Lower and Middle Ordovician Whiterockian Series: Bulletins of North American Paleontology, v. 369, p. 33–50.

Tucker, R.D., and McKerrow, W.S., 1995, Early Paleozoic chronology: A review in light of new U-Pb zircon ages from Newfoundland and Britain: Canadian Journal of Earth Sciences, v. 32, p. 368–379.

Tucker, R.D., Bradley, D.C., Ver Straeten, C.A., Harris, C.A., Ebert, A.G., and McCutchen, S.R., 1998, New U-Pb zircon ages and the duration and division of Devonian time: Earth and Planetary Science Letters, v. 158, p. 175–186, doi: 10.1016/S0012-821X(98)00050-8.

Tucker, R.D., Krogh, T.E., Ross, R.J., and Williams, S.H., 1990, Time calibration by high precision U-Pb zircon dating of interstratified volcanic ashes in the Ordovician and Lower Silurian stratotypes of Britain: Earth and Planetary Science Letters, v. 100, p. 51–58, doi: 10.1016/0012-821X(90)90175-W.

Vandenberg, A.H.M., and Cooper, R.A., 1992, The Ordovician graptolite sequence of Australasia: Alcheringa: An Australasian Journal of Palaeontology, v. 16, p. 33–85, doi: 10.1080/03115519208619032.

Van Hinte, J.E., 1976, A Jurassic time scale: American Association of Petroleum Geologists Bulletin, v. 60, p. 489–497.

Westermann, G., 1984, Gauging the duration of stages: A new approach to the Jurassic: Episodes, v. 9, p. 26–28.

Williams, S.H., and Stevens, R.K., 1988, Early Ordovician (Arenig) graptolites of the Cow Head Group, western Newfoundland, Canada: Palaeontographica Canadiana, v. 5, p. 167.

Young, G.C., and Laurie, J.R., 1996, An Australian Phanerozoic timescale: Melbourne, Oxford University Press, 279 p.

Zalasiewicz, J.A., 1990, Silurian graptolite biostratigraphy of the Welsh Basin: The Journal of the Geological Society, London, v. 147, p. 619–622, doi: 10.1144/gsjgs.147.4.0619.

Ziegler, W., 1978, Devonian, *in* Cohee, G.V., Glaessner, M.F., and Hedberg, H.D., eds., Contributions to the geologic time scale: American Association of Petroleum Geologists Studies in Geology, v. 6, p. 337–339.

Manuscript Received 31 October 2007
Revised Manuscript Received 14 July 2008
Manuscript Accepted 25 July 2008

Printed in the USA